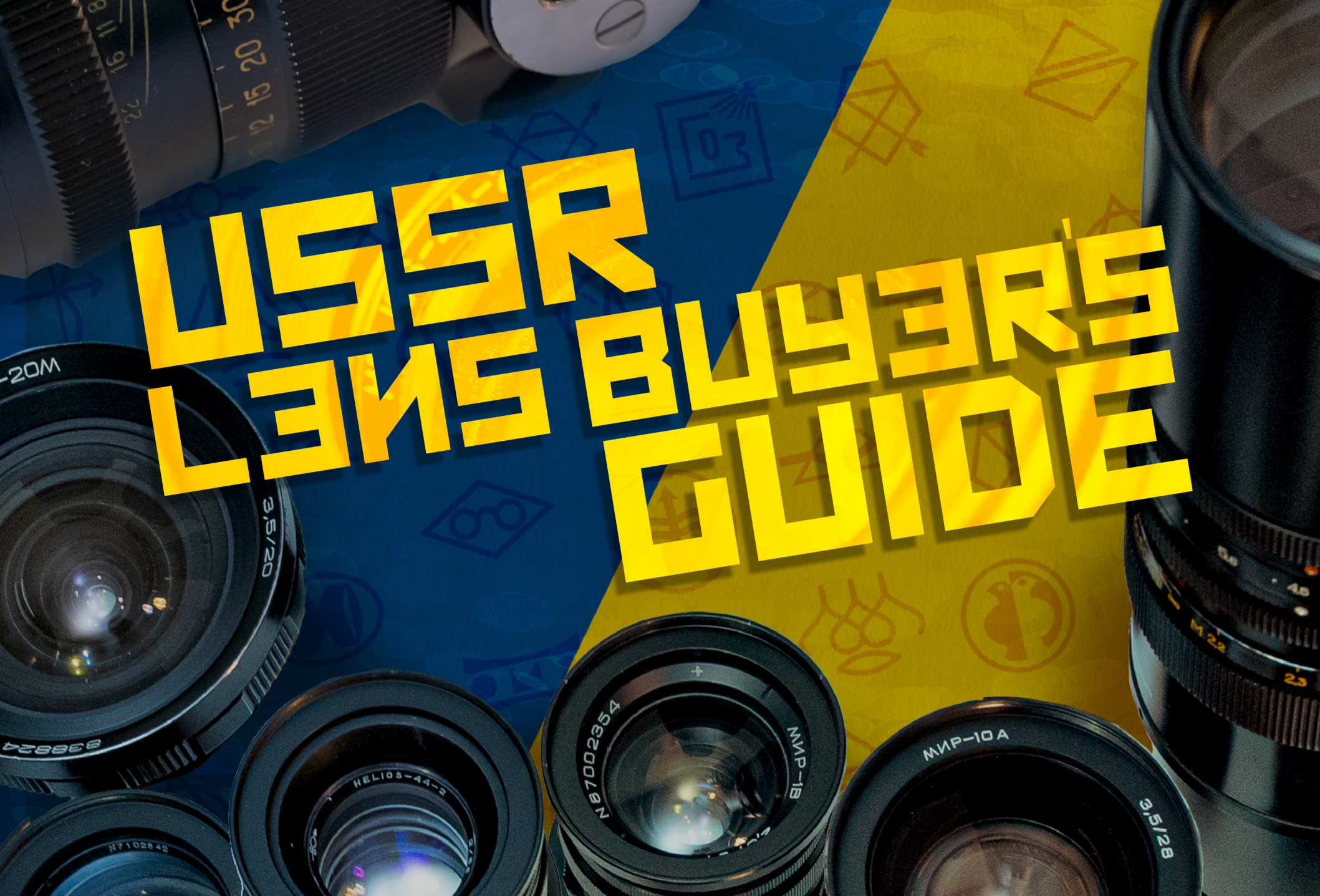
USSR
LENS BUYERS
GUIDE

USSR LENS BUYER'S GUIDE

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1. INTRODUCTION

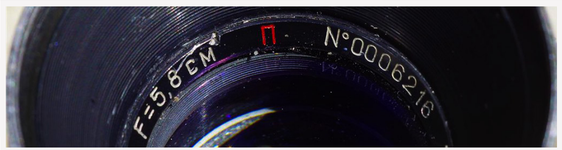
The goal of this guide is to help all photographers and cinematographers in their quest for vintage USSR - or Russian, or Former Soviet Union (FSU) - lenses. Our journey starts with *"why are these lenses so popular in today's super-high resolution sensors and ultra high definition videos?"*. We will look at what are the reasons for when asking *"which vintage lens should I buy?"* at least one of the answers is *"Helios 44!"* and probably other Soviet lenses pop up among the suggestions.

The famous Helios 44. You will have at least one of these.

My selection of lenses for this guide is based on the most adaptable mounts (M39/M42) so you can fit them to any cameras, DSLR or mirrorless, from small sensors all the way up to full frame.

Helios 44-2 adapted to Sony full frame camera.

Then we move to some of USSR's lenses' technical aspects and what are some of their distinctive features. The fact they intercut so well among each other makes it easy to put a set together. I will teach you how to quickly identify focal length, aperture, brand, model, version and mount just from reading a lens name. The next subject is the controversial red Π and a little bit about the early versions of USSR lenses - including a quick note on how to identify some more desirable models based on serial numbers.



The controversial red П.

The $350 Starter Set chapter shows you how cheap it can be to assemble a standard set with three focal lengths. If you are in a rush to grab some glass and get filming before reading the whole guide, this is where you should start.

A collection of early silver models, by Vintage Lenses for Video.

After this general information on how the Soviets worked their optics, we get to an extensive list of the most popular and respected twenty lenses from their ranks. These are lenses you can get on eBay right now for cheap. I am talking tech specs, rumors, different versions, coatings... You will find valuable information to help you make a choice when buying USSR glass, be your mission a one-time thing or a full-blown project of building the very best set anyone can think of (you will have to fight with me for that title).

After the lens list, you will find a chapter on rare and exotic lenses that are nearly impossible to find so you can stay up for months trying to collect them. Then we move onto a series of short chapters with useful bits and pieces of information such as recommended and knowledgeable sellers; disassembly guides and modification instructions; or where to go to service USSR optics. At the very end I briefly explore Zenit's new-ish fast primes which promised a revival of Soviet's classics. Yet, did they deliver it?

Zenit's new-ish super fast 50mm f/0.95.

Besides all the hard knowledge, this guide aims to provide you with some entertainment. During my research I came across unbelievable myths about these optics and how they were treated in the USSR. So, if something sounds absurd, there is a chance it is just an internet rumor! Take those bits with a grain of salt!

Shall we start our journey, comrade?

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2. REASONS AND RULES

When I first started in photography, it was all about getting the best lenses. Faster apertures, lightning-fast autofocus, zoom lenses to cover all focal lengths, constant aperture, the whole nine yards... In fact, the less artifacts the better: less flares, less distortion, better light transmission, sharper results. L was the way to go. It filled all my dreams and for a while I had quite the collection.

Canon EF 24-70mm f/2.8L, 135mm f/2L, 85mm f/1.2L, 35mm f/1.4L. I had these four and many, many more.

I started to change my mind when I got into film school in 2010. Autofocus was rendered useless as I learned there was a person whose job is to getting images in focus (1st AC).

Left: First year of film school. "What is camera?". Right: Third year of film school. "Ooh!".

As projects became more elaborate, exposure adjustments during a shot were required and things get jumpy when aperture is controlled electronically. The camera snaps the lens aperture to the default values in a heartbeat and just as quickly ruins a smooth, fluid shot with a jarring exposure click. It is a two-step problem, first, the camera controls the aperture, and second, aperture follows the standard values (1.4, 2, 2.8, 4, 5.6, 8...) with sharp clicks even when the lens has an aperture ring.

Markings on a clicked aperture ring.

By the end of my film program I had fallen out of love with modern lenses - not that I was able to say this so clearly back then. No matter how much post-processing I threw at the footage, everything came out looking blandly similar. It looked just like anything else. It lacked personality.

My graduation work was about anamorphic adapters and one of the first things you read about anamorphics is that they favor lenses with simpler optical designs. It did not take long before I started to look into Soviet lenses and other vintage optics.

Helios 44-2 and Kowa for Bell & Howell Anamorphic Adapter.

For specific reasons (I admit: lens flares) I jumped head on into Soviet lenses instead of German or Japanese optics, putting together a set that would get the best out of my anamorphics. Since 2012 I have had countless Soviet lenses - I would risk saying over a hundred - and I currently own a set of primes that range from 20mm to 500mm. What still holds my interest about them is their almost endless variety when compared to other brands and the organic and imperfect look they deliver.

Lens whacking with a Helios 44 for extra organic bits. The Emotional Octopus, 2018.

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Since a full list of all Soviet lenses is an impossible task, for the purpose of this guide I will focus on a smaller selection, picked based on careful criteria to be explained in a moment. This smaller selection also allows me to add depth about each model, not only the basic information that you can get with a Google search. These are the best, the most popular, the cheapest, the legendary, the unique. These are the lenses that make people love and desire Soviet vintage optics.

Although tempting, calling these lenses Russians is not entirely accurate. Most of them were in production during late 1940's to late 80's, not in Russia per se, but in the Soviet Union. The USSR was one of the world's most powerful nations at the time, going head to head against the United States in terms of scientific and military advances.

Map of the Soviet Union by R. M. Chapin.

There was an economic and ideological race on and technology jumped leaps because of it, including optics design and manufacture. So, even though these lenses are abundant and cheap today, they were quite expensive in the past - a person would need over half a month's salary to afford a Mir-1 -, made by great scientists and designers. They are not some cheap knock-off with low quality materials, even though there certainly are low-quality versions out there!

Lens assembly line at KMZ optical plant, 1956.

My main criteria for choosing the lenses featured here is adaptability. There were plenty of lens mounts and camera systems being used in the USSR between the 1940's to the late 1980's. Not all of these systems are adapter-friendly. Even though many of us are moving - or have moved - towards mirrorless camera systems, there is still a large audience using reflex cameras alongside cinema cameras and their PL mounts.

Rehoused Helios 44-2 on RED Komodo camera. By p_x_c.

For that reason I chose M42 mount as the common denominator. M42 is great for its ease to fit other camera mounts. It was also the most produced mount during the 1940-1980 stretch, which leads to an almost endless supply to our demand. There might be a few M39 lenses in there, but these are Zenit-mount, not Rangefinder LTM.

Leica Rangefinder M39 or LTM cameras and lenses. You do not want this M39 mount for Soviet lenses.

The M39 Zenit lenses need a M39 to M42 ring and a 0.3mm spacer, then you are good to go. M42 adapters for other mounts are super cheap and readily available, so it is easy to have one adapter per lens, instead of constantly switching the adapter from one lens to another.

A Helios 44-2 with adjustable M42 adapter: the markings are readable from the side.

If you are particularly picky (I am) about which direction the markings on the lens are facing, I highly recommend these better built, custom rotation adapters. With them you can quickly realign the lens with the mount so the markings face upwards or towards your 1st AC.

I also recommend to steer clear of adapters with chips of any sort (AF confirm, more commonly), and rely on simple mechanical adapters. I have had bad experiences with chipped adapters in the past, plus they do not add any value for my work.

As a disclaimer: all eBay links are going to be highlighted red – and if you buy from any of these eBay links, I get a small commission, so help me out! -. Regular links are blue, and this is not a link... Sorry.

3. CHARACTER OR PERSONALITY

There is a recurring saying among the people who grew tired of modern lenses and made the switch for vintage optics: *"modern glass is too clinical... vintage glass has soooo much more character, you know?"*. Honestly, that is a great way to put it when you are getting started, but after you are knee deep into vintage lenses, saying a lens *"has more character"* does not cut it. What does "character" even mean? So I propose a distinction between character and personality.

What most people try to convey by opposing the words "clinical" and "character" is that modern lenses are "efficient and unemotional; coldly detached" (Google: "clinical"). They simply serve their purpose of capturing images, as transparently as they can. Today's level of lens design and production are eons more advanced when compared to the way optics were planned and crafted just a few decades ago. Computers and machines are much more capable of achieving technical perfection than us humans, so all modern lenses are as perfect as they can, with a few trade-offs for size, weight and especially final price. This perfection in design and manufacture is what leads to the clinical look. In the comparison on the next page you can see the left side of the frame has smoother bokeh, better brightness and sharpness, plus neutral colors.

Match framing and composition: Canon EF 50mm f/1.4 (left) vs Helios 44 (right).

Vintage optics, conversely, are riddled with imperfections. The combination of these imperfections is what we call "character". The list to pick these imperfections from is long: distortion, chromatic aberration, astigmatism, spherical aberration, various types of flares, vignetting, softness/sharpness, unusual bokeh, different coatings, coating damage, bad quality control, wear and tear, modifications, scratches, dust, fungus, and so on. Looking at these perks, we can say "character" just means the lens is far from optical perfection.

A problem arises because **every lens has character in the sense that no lens is actually perfect**, not even modern ones. Modern lenses are closer to perfection, yes, but not there yet. Whenever someone says a lens has character, they mean it in a good way. I never heard of anyone saying *"I hate that lens because it has too much character"*. This is why I feel the urge to say USSR lenses have not only character, but personality.

I argue USSR lenses have their own personality because they will not just do whatever magic you expect them to do because you want to see it. Being capable of rendering amazing images does not warrant that anything captured with them will look amazing. There are plenty of boring/bad shots taken with USSR lenses, meaning you have to learn how to handle them, share experiences together, test the crap out of everything and come up with your own favorite tricks and techniques.

These are USSR lenses of a higher tier: LOMO Cinema lenses!

Having character is awesome, but having a personality means they might act in certain ways in different situations. For example, the Helios' swirly bokeh is one of its main traits, but the swirl will not show up in every single photo/shot you take with the Helios. It needs a specific mix of background elements, lighting conditions and distance between photographer, subject and background.

Swirly bokeh from a Helios 40. Photo by Mathieu Stern.

Besides their "model" personalities, almost every lens is different from its siblings even if they share the same name and year of production. This is particularly easy to notice among several Helios 44, yet it can be said for any other USSR lens. One copy flares more easily, another has superb contrast while the third one is totally washed out, then the fourth is softer than all the previous ones, and so forth. A popular internet belief is these discrepancies come from leniency in the quality control department of USSR optical plants, so all these variations were acceptable.

This theory has been questioned again and again by Soviet lens experts such as Ilya Volkov from Moscow Optical Workshop.

Volkov argues this is a myth and that in fact quality control was superb then, but many of these lenses have been so poorly taken care of over the years that they could all benefit from servicing. He is also responsible for compiling the Russian | Soviet | USSR Lens Survival Guide on REDUser, providing tons of extra information on the matter.

Besides this, the USSR was gigantic in geographic terms and they had a number of different production plants with their own workers, tools and methods of delivering the same optical formula, meaning the outcome would always vary slightly!

Unlike other brands - Zeiss, Nikon, Canon, you know -, which have their exclusive factories, each optical plant in the USSR made lenses for several brands. We tend to think of USSR optics as a single family, one unique brand, but it turns out that is incorrect. The words we like to think as the names of the lenses are actually their brands, and their models are represented just by the numeric part. For example, Helios is a brand, some of its models are 44, 77M, 103, 40-2, Jupiter is a different brand, famous for its models 9, 21M, 37A. That applies for all other lenses.

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You better believe all of these are different USSR Helios lenses.

There was an overlap of brands, models and optical plants. If we look at the classic Helios 44, a copy could have been produced by Valdai, or KMZ, or MMZ. The same would happen to a variety of other lenses and other plants.

These were the main factories for USSR optics.

During World War II, KMZ was the USSR's main optical factory. They received *"all of the captured German optical data and lens manufacturing equipment. This caused them to become the primary optical design and development facility in the Soviet Union (...) They developed designs for their own production and built initial runs of others before transferring production to other factories"* according to Communist Cameras. So their lenses should be more consistent or better built than the rest - but remember: bad copies can be found anywhere, it is just a matter of luck.

Greetings from the Opticians' Square, in Krasnogorsk!

4. TECHNICALITIES

In this chapter we will go over a few unusual traits and overall technical aspects regarding USSR optics. We start with Preset Aperture, there is an explanation on how it works today and its advantages over regular aperture rings as well as how it used to work back in the day. The second part of this chapter covers naming conventions, what were the codes followed by Soviet lens manufacturers when defining each lens' name. By learning those you can quickly grasp a number of things about an unknown lens just by reading its name. The last bit is about the many common qualities that span over USSR lenses making them great matching sets. These optical qualities give your footage an edge of personality over modern super-sharp glass and random vintage picks.

4A - PRESET APERTURE

One thing that sets some Soviet lenses apart from regular vintage lenses is their preset aperture. Preset aperture was the standard aperture control before the late 1960's, particularly common in M42 lenses.

The preset aperture rings on the Jupiter 9 lens.

You know the DOF button every digital camera has? It engages the aperture blades in photo mode, so you can see how the depth of field for your photo is going to look. If the DOF button is not pressed, the camera holds the aperture wide open for more light to go through and make it easier to focus, or even just to see through the viewfinder. In video mode that does not happen, and the aperture blades are always engaged.

An electronic aperture's blades only engage when taking the photo.

When adapting lenses from different mounts, a common issue comes from electronic-controlled apertures. In these lenses the iris is fully controlled by the camera body, lacking an aperture ring. This type of control renders many of these lenses - notably Canon EF - unusable for video work across different cameras, and leads to the need for "smart adapters". These adapters read the electronic signals coming from the camera body (a Sony A7sIII for example), and translate them into the lens' language, allowing for a Sony camera to control an EF lens. That is the most extreme case of adapters.

Sony E-mount camera, Canon EF lens. EF to E-mount adapter in between them. Notice the electronic contacts inside the mount.

If we go back in time a little bit and look at lenses with manual aperture control, the aperture ring is usually at the back, near the camera body and features f-stop numbers written on it. It rotates in defined steps between these numbers, with a pronounced click every time you open or close the iris. These are nicer for video work than an electronic aperture, since you can adjust the aperture in the middle of a shot without changing anything in the camera, just touching the lens.

See the aperture ring near the camera body, under "Contax". Photo by Casual Photophile.

The problem is these clicks make the exposure jump whenever it is changed. The use of photography lenses for video work has created a demand for a process called de-clicking, which is essentially removing the small bearing that lives under the aperture ring which makes the stops click. Declicking a lens will give you an aperture ring that allows smooth aperture changes during a shot.

Prior to clicked aperture rings, there was preset aperture, which might seem counterintuitive for our modern minds, but it was a nice – click-less - way to control the aperture. First of all, lenses that have preset apertures have two aperture rings and they are located near the front of the lens.

There is a red dot on the rim of the filter thread and below it is the numbered aperture ring. This ring is used to pre-set the aperture to your desired final f-stop and it clicks as you change the values, but you will notice that this alone will not engage the aperture blades. Below this first ring is a second ring, much softer. This is the ring that actually controls the blades, so rotating it counter clockwise will close the iris up to your preset value - when the second ring hits a hard stop. Rotating clockwise will open it back all the way. This type of control allowed photographers to choose their exposure and quickly switch between an open iris for a brighter focusing screen, and back to the desired value right before pressing the shutter as explained on Pentax Forums.

Preset apertures let us, modern video folks, define our most closed aperture value and smoothly transition between that and wide open, adjusting to the external lighting conditions. I usually leave the preset ring at the smallest aperture (f/16 or f/22) and just use the second ring to adjust exposure according to what I see on the screen/histogram.

Preset is as good as a declicked ring, with the benefit of not costing anything extra since it is already there as a part of the lens design. Many USSR lenses have these rings and I miss this feature being mentioned frequently, so, here it is: another advantage of going Soviet. Its biggest drawback is being challenging to tell what exact f-stop you are at.

48 - DIFFERENT NAMES, DIFFERENT DESIGNS

Sticking to the Helios 44 as an example: there are plenty of different versions of it. Each one has a different suffix to its name. There is a 44-2, 44-3, 44M, 44M-7, and many others, besides a plain Helios 44 with no suffix. Whenever the lens design was changed or updated, the Soviets increased that last number. This design numbering was a common practice among USSR optics and it applies even to some cinema lenses which I will not cover here. Needless to say, whenever they updated the version number, it usually meant the lens had been improved optically. The early Helios 44 models have a resolution of 35/14 lines per mm at center/edge while the later 44M-7 resolves 50/30.

A wide variety of Helios lenses with slightly different names.

Not only the Soviets used these numbers to keep track of their versions, there are several letters that also have their own meanings regarding mounts. "K" stands for Pentax K; "M" for Modified, which means an aperture pusher pin on rear side of the lens. "M" lenses are always M42; "N" for Nikon F; "A" means it is a swappable mount, so "A" lenses can be used on both M42 or Nikon F systems with full range focusing. You just need to get the appropriate back and switch the mounts; "T" for T mount (or TV in some exotic cases) ; "S" for sniper (!!).

Some of the lenses have versions with no letters at all - the early ones -, and versions with different letters when they started to be made for different mounts. All of these different names build up on that note of different lenses made in various optical plants that are actually the same design and brand, adding exponentially more slight differences between them.

The general formula for lens naming is:

BRAND #X-Y

BRAND is obvious: Helios, Jupiter, Tair, Industar...;
 # stands for the model (44, 40, 11, 20);
X is the mount;
Y is the version/design.

Sometimes not all of these parts show on the name, like the first versions of the Helios 44 do not mention the mount, so it is good to understand their naming convention, but check thoroughly whenever buying earlier models.

Helios 40, Helios 44, Mir 1 in early silver versions.

I got a couple more notes about the earliest models. Soviet lenses start their serial numbers with the last two digits of their year of manufacture. A Helios 44-2 with serial 7321748 was made in 1973. This applies to any USSR lens. Once you start to go through them you will notice a few that do not follow this rule and start with zeros. The explanation for these is found at a MF Lenses post, and is a bit of internet legend to influence prices. Take it with more than a grain of salt.

000 - factory test models (prototypes)
00 - pre-series, reserved for members of the communist party and VIPs
0 - reserved for factory union leaders

On the same post, the author states that this information came from Alain Berry, an expert with connections with Soviet lens' manufacturers and these lenses "*are in fact all pre-production models. They are generally considered higher quality optics due to the fact that there is stricter quality control (...), they are produced in lower numbers with greater care, and they are not presented to the general public. A great many members, with multiple copies of the Helios 44-2, have invariably found the best to be the one with the 0xxxxx serial number*".

Since we are talking about the Helios 44, the multiple zeros only apply to the earliest, silver versions of the lens. According to internet history, the MMZ optical plant scrambled their serial numbers during the 1970's, meaning plenty of lenses came with serials starting on 0's, when they should start with 7's. Lastly, a few other USSR lenses were manufactured well into the 2000's, so their serial might start with zeros. If the lens looks too modern, it is not a pre-series.

Whether this is true or not, I still went after a 00 lens set and have no regrets except that I probably overpaid for every single lens!

- This bit about the serials is 100% an internet myth, confirmed by Roman Bibik, at Retro Foto House.

4C - THE RED П (P)

As soon as you get into looking for specific lenses and versions, you will notice more and more people (and sellers) mentioning a special "red P" - **P** is the roman equivalent of **П** from the cyrillic alphabet. Some of the earliest USSR lenses had this making on their front ring. It was used to represent special coatings just like Zeiss' T coatings. Some people say these versions are better than the ones that came after them, for they are closer to Zeiss' original versions and even go as far as saying that these used Schott glass taken from Zeiss' production plant in Germany after the war, as reparations for the damages and casualties suffered by the USSR. Others state that this was just a mark meaning that the lenses had some sort of coating, and then when coated lenses became the norm, they just ditched the symbol circa 1960.

One thing is certain about the lenses that have this marking though: their coating is different when compared to other versions. If that is a good or a bad thing, it depends on how you like your flares. Coatings continued to change over time, so even among lenses without the **П** coatings can be different. For a more detailed read, this article from Soviet Photo explains the early stages of introducing multi-coating (MC) to USSR optics, including why some coatings are colder and others warmer, and the consequences MC brought to the images.

Mir 20M and smooth coatings. No red П though.

4D - MATCHING OPTICAL QUALITIES

One of the main reasons people like and go for USSR lenses is because they are cheap and their look matches nicely across an extensive variety of primes. Most people with matching sets knew exactly the lenses they were looking for, down to the serial number because the year of production has a direct link with the coatings and design, affecting the overall look of each lens. Some more obsessed photographers and cinematographers even get lenses from the same optical plant so their imperfections are also similar.

Helios 44 (top) and Mir 1B (bottom) matching tone and colors.

I could not find much information on how exactly the coatings changed throughout the years but, in theory, similarly colored coatings should result in flares within the same ballpark. If you get an early copy of the Mir 1, with blue/purple coatings, that will very likely not match a neutral coated Helios 44-2. Also, lenses in the same brand tend to have other matching qualities and general design, such as the swirly bokeh of the Helios 44 (58mm) which can also be found in the Helios 40 (85mm).

It is subtle, but the top image is warmer, greener, and contrastier.

In terms of color, USSR glass swings to warmer tones; oranges and yellows rather than blues or greens because of their blue-ish coatings, especially in the earlier lenses. This warm feel makes the images look cozier and more pleasing to the eye than a color-accurate rendition. I would advise against correcting this color shift through white balancing on camera since the warmth is not coming from the light itself, but from its interaction with the glass surfaces of the lens.

Many people using vintage glass are also fans of the lowered contrast introduced by older coatings. The overall image
is washed out when compared to any modern lens as coatings were not so effective and light-transmitting then as they are today. When you have light hitting the lens directly, that is when you really see how vintage and modern glass are different. Any strong or direct light source in the frame is likely to have a halo around it, wrecking contrast. This sometimes affects only the light area, but sometimes it washes out the entire frame. The lowered contrast varies from lens to lens and version to version, so experimenting and research are key.

5D3 + Jupiter 9 + Kowa for Bell & Howell anamorphic adapter.

The way USSR lenses handle flaring ties back to the idea that they have their own personality. Some of my Helios flare way more when the light is just outside the frame rather than in the middle of it, while others just wash it all out in white. Flares are also not restricted to washed out blooming highlights. Many Soviet optics are famous for how their elements show up in the footage against strong lighting. The Helios 44 might be a strong contender for crazy flares, but the Mir 24H's also comes up along with others lenses from this era. Their flares are not identical, but they have some unity. You can intercut footage shot with any USSR lenses and the image will more easily match among them, unlike trying to cut from a Helios 44 to a Canon EF 85mm.

Glowing highlights with pronounced chromatic aberration.

Notice the blooming of those highlights over the subject's dark hair.

Helios 44 with an almost direct light hit.

It is a consensus that none of these lenses are razor sharp - USSR lenses are notably soft wide open and if sharpness is key for you, these are probably not your thing.

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Their overall soft texture is one of their key features over modern glass. Soft texture does not mean blurry results or soft focus. Sharpening a little bit in post works like a charm. As Ken Rockwell wisely puts it, today we are taught by lens manufacturers to obsess about sharpness and edge to edge performance. Sharpness has become a marketing move to promote "better" lenses. This kind of thinking takes creatives further and further away from the right idea of what lenses should be doing - with character and personality. Stopping down any USSR lens will yield great, sharp enough results on any camera.

Give me chromatic aberration, edge blurring and vignetting! I love this look.

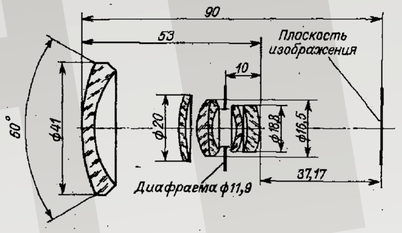
5. THE $350 STARTER SET 37, 58, 85MM

I wanted to add a chapter to this guide for the people who would not have the time or patience to read the whole thing. The goal here is that you can quickly get enough grasp to build your own matching set in a few simple steps. As the title states, it is a cheap set, costing a fraction of what a single new lens costs nowadays - with this entire set you can afford half of a Canon EF 50mm f/1.4. These three primes also cover the standard focal lengths for video, giving you a wide angle, a medium lens and a telephoto.

The Starter Set and a Tair 11A trying to intrude in the background. You will read more about it soon.

It is no coincidence these lenses are famous for their results. All three of them are copies from Zeiss' blueprints obtained by the USSR after World War II and adapted into USSR's own designs. By getting these three lenses you are basically getting Zeiss designs for a fraction of their price and much more added personality. This trio is not the fastest nor the cheapest of USSR lenses - can you believe it?!, but they have by far the best performance/price ratio.

37mm - Mir 1B

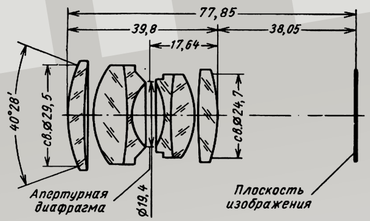


Mir 1 optical diagram.

The 35mm, or our wide angle, is actually a 37mm, f/2.8 **Mir-1B**. Price ranges between $80 and $100 and it is an improved version of the Zeiss 35mm f/2.8 Flektogon - with plenty of detailed comparisons online. The Mir has very little distortion and chromatic aberration even though it is prone to flaring. Being such a tiny and lightweight lens, it is no wonder people get attached to it.

A great choice for wide and establishing shots. Photos by Yulia Rusanova.

58mm - Helios 44-2



Helios 44 optical diagram.

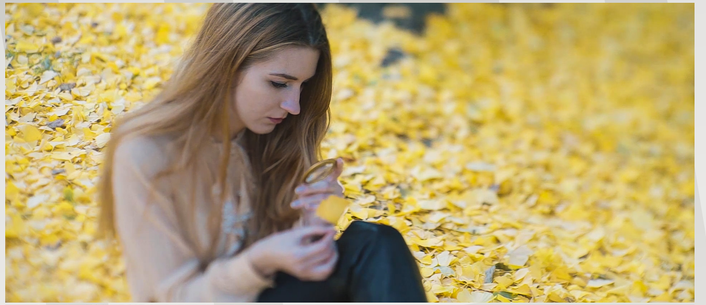
For 50mm, the classic **Helios 44**, 58mm f/2. It usually goes from $30 to $80, and ships from eastern Europe. The Helios 44 uses a Double Gauss design, identical to the Zeiss Biotar, which gives this lens its most memorable feature: the swirly bokeh that shows up as you move towards the edges of the frame. Not only that, it is a design with almost no distortion and little chromatic aberration. The Helios 44 has many different versions and each person has their favorite. Mine is the 44-2.

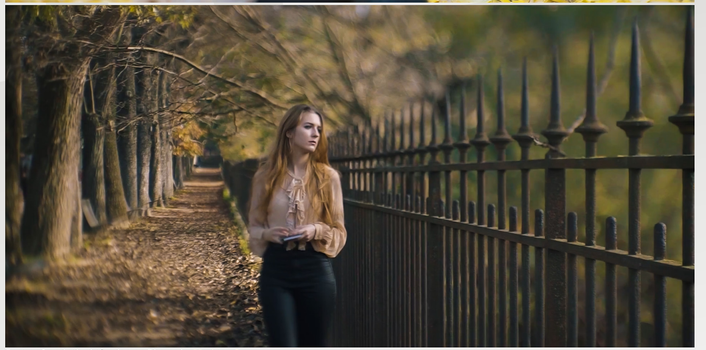
Photo by Julia Trotti.

85mm - Jupiter 9

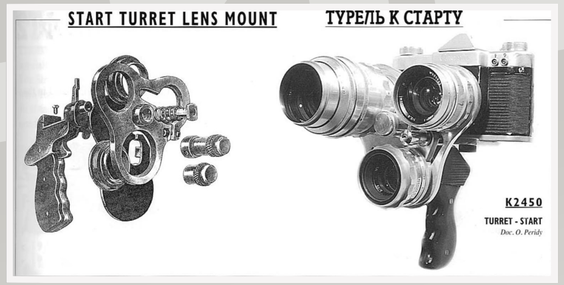
Jupiter 9 optical diagram.

The **Jupiter 9** is the 85mm telephoto of choice, and this time it is really 85mm, f/2. The Jupiter is the most expensive lens in the set, going from $200 to $240. It is a copy from Zeiss' Sonnar design, famous for its fast aperture and sharp results. Like the other two lenses above, focus throw is long, almost a full turn, and, if you are getting the earliest versions featuring preset aperture, no clicks to ruin your exposure adjustments.





Jupiter 9 and tilt shift adapter. Photos by Mathieu Stern.



It would be awesome to mount this starter set on a lens turret.

Most USSR lenses ship from eastern Europe (Russia, Ukraine, Poland, Lithuania, Latvia) and delivery times are long (at least three weeks to North America). We learn two things from this: the first is that most buyers of vintage USSR optics do not sell their lenses, they just keep them - which is a good sign. The second is supply is so vast that you can rely on prices being almost constant. There will always be a few listings off the curve for extremely cheap (my first Helios 44-2 costed $8, including shipping), but the average prevails. A constant supply also means that you can choose your versions carefully, for most of them are common enough to be always available. By being able to quickly find your selection, it will not take you months of eBay hunting for any specific lens.

The goal of this set is that you can go on eBay NOW and get all three lenses for around $350. Click here for the Mir 1, here for the Helios 44 and here for the Jupiter 9 if you still do not believe me about the prices, and get yourself a set!

6. THE LENSES

In this chapter I am quickly reviewing and providing technical information on almost twenty lenses: the most popular choices among USSR glass for modern photographers and cinematographers. I took into account availability, price range, performance and personality. Each lens has a pretty profile picture, technical specs, average price (as of mid-2022), a bit of history and useful information for choosing your copy.

BRANDS

A quick overview the brands and what to expect as to overall characteristics from lenses within each category.

HELIOS (Гелиос)

Even though Helios has a wide variety of models, most of their focal lengths in M42 mount are around 50mm. They are what you would call a thrifty fifty: a medium (as in "not wide angle nor telephoto") fast lens that came with cheaper Zenit and Zorki cameras. The most famous models are the Helios 44 and 40, with focal lengths of 58 and 85mm respectively. Because they follow the Zeiss Biotar design that gives them, under the right conditions, swirly bokeh.

JUPITER (Юпитер)

The Jupiter lenses for M42 are geared towards the telephoto side, with most of their models over the 85mm focal length. Many of them derive from the Zeiss Sonnar design. The name of the family - Jupiter - comes a trait of its optical design: a three (or more)-group construction with one element that is considerably thicker than the other elements, according to Camerapedia.

MIR (Мир)

Just like the Jupiter are mostly telephoto, Mir (which translates to "world") are wide angle lenses. They range from 20 to 37mm. Mir's are not as fast as Helios and Jupiter (f/2.8 or slower compared to f/2), but introduce very little distortion for wide angles. Users also tend to praise the look of their flares.

TAIR (Таир)

The Tair brand covered long telephoto lenses – the shortest one is 135mm, all the way up to 500mm - and here I will cover the two M42 models released: the Tair 11 and 3. The Tair 3 is popularly known as the Photosniper, an unusual camera rig that resembles a sniper rifle with a super long lens. Needless to say I do not recommend using such rig these days.

INDUSTAR (Индустар)

Just like the Helios, Industar was a brand of cheap kit lenses that came with the camera bodies. Slower when compared to the Helios - f/2.8 or slower as opposed to f/2 -, these tend to go cheaper. They are also more compact and lighter than the Helios. Based on the Tessar design, Camerapedia says "*the brand name was a classification for certain 4-element 3-group lens constructions*" which points to the fact that even though the various Industar models are different lenses, their core is the same.

OTHERS

A few lenses reviewed did not fit in any of the previous groups, such as the MTO mirror lenses for longer focal lengths with compact bodies. Besides them, there are still plenty of other USSR brands that will fit smaller sensors or shorter flanges. These lenses will not be covered in this guide.

USSR LENS BUYER'S GUIDE

Twelve of these lenses will be covered in the next pages, plus a few not shown. Photo by Siim Vahur.

HELIOS 44

HELIOS 44

Focal length: **58mm**
Aperture: **f/2-16/22**
Aperture Blades: **6, 8 or 13**
Minimum Focus: **0.5m**
Focus Throw: **270° / 180°**
Filter Thread: **49 / 52mm**
Weight: **230/270g**
Length: **57mm**
Avg. Price: **$30-80**

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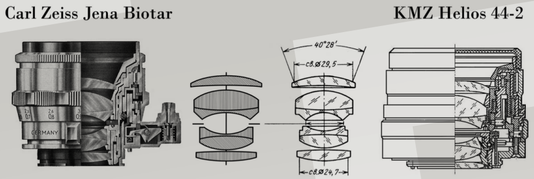
Produced in countless numbers since the late 50's and all the way into the late 90's, the Helios 44 is one of the introductory lenses to USSR optics. They are so easy to find and simple to pick apart, regardless of version, that photographers do not feel afraid to experiment with them.



A comparison between cross sections and designs of the Zeiss Biotar and the Helios 44.

It is a fast standard prime that came with many Zenit cameras, meaning it was designed for full frame. Being a copy of Zeiss' Biotar design ensured the Helios 44 had excellent performance even though being crazy cheap nowadays. One of its most famous personality traits is the swirly bokeh that gives a great sense of motion to the edges of the frame under the right circumstances.

Initially made for the internal USSR market, some Helios 44 have inscriptions in Cyrillic letters, while others have it in Roman letters. There are rumors the Roman ones were made as export versions, but there is no conclusion if these were better or worse than the Cyrillic versions!

Throughout its production run, the design was updated and changed a number of times. The earliest versions (Helios 44) featured a M39 mount, had a silver body, inscriptions were in Cyrillic, focal length was in centimeters (5,8cm), and the minimum aperture was f/22, with 13 blades. These have serials starting with #000 and the desired **П**. The next versions still had 13 blades and silver body, but aperture was limited to f/16, serials start with #00 and also feature the **П** for unique coatings. A little before the black body became a standard (late 60's) aperture blades were reduced to 8.

Canon 5D3 + Helios 44-2 + Iscorama 36 – swirly bokeh and AK-47!

My favorite models are the earliest ones (44, or 44-2), for I had better luck regarding quality, plus I like the preset aperture (clickless iris adjustments) and lots of flaring. Coatings were improved on the later versions (44M-4 and above, from 1979) and aperture blades were reduced to 6, making round bokeh impossible to achieve when stopping the lens down.

On the bright side, resolution improves sensibly across models, starting at 35/14 lines per mm, center/edge, on the original Helios 44 up to 50/30 on the Helios 44M-7. With the change to the M models, filter threads changed from 49mm to 52mm and focus throw got shorter, from 270° to 180°. Minimum focus changes a little bit from version to version, but it never falls too far from 0.5m / 1.5ft.

Swirly close ups. Photo by KeenanRIVALS.

WARNING! Beware of the Helios 44-3 model, since many of them are not as adaptable to EF, in spite of being M42 mounted. Another note is the earliest Helios 44 have non-standard filter threads, M49x0.5 instead of M49x0.75, requiring specific adapters. RafCamera makes them upon request.

If you want more detailed information about specific models and factories, I recommend checking this list focused exclusively on Helios 44's tech specs, production plants, serial numbers and versions. Or if you just want to know which specific model to look for, watch Retro Foto House's Ultimate Buying Guide on YouTube where Roman Bibik goes over different manufacturing plants and identifies common problems good performers among many different Helios 44-2's.

HELIOS 40

HELIOS 40

Focal length: **85mm**
Aperture: **f/1.5-22**
Aperture Blades: **10**
Minimum Focus: **0.8m**
Focus Throw: **240°**
Filter Thread: **67mm**
Weight: **880/950g**
Length: **95mm**
Avg. Price: **$330-500**

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The Helios 40 is the bigger brother of the 44. With a focal length of 85mm, it still follows the same Zeiss Biotar design: a double gauss lens. Photographer and reviewer Bob Atkinks informs us that *"double gauss lenses have been used since the 1930s for fast semi-wide to short-telephoto lenses"*. Almost one stop faster than the Helios 44, the Helios 40 presents the same personality with captivating swirly bokeh.

Helios 40-2 on a RED Raven. Photo by omitVOID.

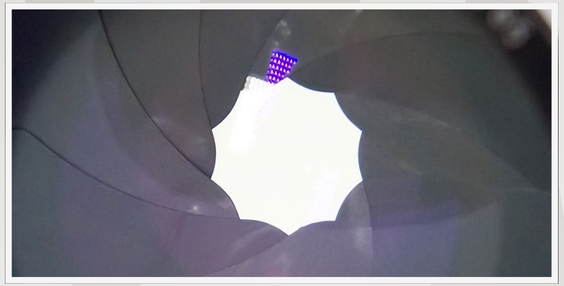
The Helios 40 comes in three versions. The original Helios 40 has a silver body and M39 mount. The earliest versions have the Π just like the Helios 44, and have better reputation for performance. They also have their front ring markings in cm (8,5cm) and non-standard filter threads (66mm), that require an adapter to work with regular filters.

Capt. Haddock demonstrates swirly bokeh, shallow DOF, low contrast, and rainbow flares.

With the mount switch from M39 to M42, came the Helios 40-2, which is always black. These were made until the early 90's and are some of the most expensive USSR lenses to this day. The Helios 40-2 featured a lens collar for tripod mounting.

All different vintage versions of the Helios 40 and 40-2. Photo by Jozef.

The third version of the Helios 40 came to be as in 2013 the lens was re-designed and released by Zenit with no change to its name - still Helios 40-2. The new version comes in Nikon, Canon and M42 mounts, introducing modern coatings. They still have the preset aperture ring. Zenit kept the build quality untouched, making a lens that is built from a solid block of metal.



Helios 40 iris shape.

Something I do not see mentioned in many reviews is how the Helios 40 displays an unusual aperture shape anywhere between wide open and stopped all the way down, with a very pronounced ninja star shape.

JUPITER 9

Focal length: **85mm**
Aperture: **f/2-16**
Aperture Blades: **15**
Minimum Focus: **0.8m**
Focus Throw: **280°**
Filter Thread: **49mm**
Weight: **380g**
Length: **66mm**
Avg. Price: **$200-240**

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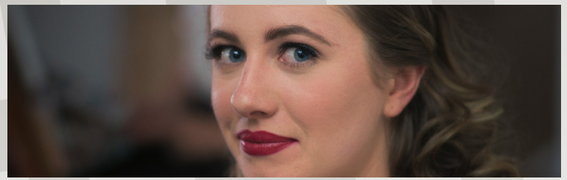
One of the most commonly recommended USSR lenses, the Jupiter 9 has plenty of features to appear in high regard. It is a short fast telephoto with 15 aperture blades that ensure circular bokeh even when stopped down. Construction is solid metal with no plastic parts. The optical design comes from Zeiss once more: the Sonnar 85mm f/2. Manufactured before WWII, its original name was ZK-86 (ZK being an acronym for Sonnar of Krasnogorsk).

The Jupiter 9 has a number of versions. Stay away from the LTM versions - see the comparison below - as they are Rangefinder lenses and will not fit any camera with a mirror.

Avoid the LTM M39 version on the left. Zenit M39 versions on the center and right.

The LTM are the first versions of the Jupiter 9, made in the early 60's – and the only ones that have the Π -, then followed by the M42 versions – first silver, then black. You can find both silver and black versions with knuckled focus rings. The newest models only come in black and have a smooth focus ring.

Besides having a smooth focus ring, all versions after 1985 have a MC inscription on the label and are reported to have softer focus when wide open. All Jupiter 9 have a bit of washed out contrast when wide open, but the newest ones are more pronounced. Detail is there, it is just contrast that is lacking.



The Jupiter 9 is not only famous for its amazing performance, but also for its wide variation in quality. There are many, many bad copies out there. For this reason, the saying goes *"if you got a bad copy, try another one, or three!"*

An important feature of the Jupiter 9 that polarizes its users are flares. When you have light coming directly into the lens, it is quite common to end up with an entirely washed out white image. There is lots of blooming and some people might hate that, so here is your warning. Blooming is also more noticeable in the newest models rather than the old ones.

This was one of the first USSR lenses I bought, because it would pair nicely with anamorphic adapters. I never regretted the purchase and it does indeed give amazing results when combined with anamorphic lenses. Its simple design is nothing but encouragement towards modifications and experiments.

Canon 5D3 + Jupiter 9 + Isco Widescreen 2000 MC anamorphic.

Some photographers consider the Jupiter 9 a soft-focus lens, especially when wide open. This feature, combined to its organic look and 85mm focal length make it a powerful yet cheap portrait lens. A long focus throw makes this lens useful for video, and its character allows for plenty of dreamy shots.

Jupiter 9 and tilt shift adapter. Photo by Mathieu Stern.

JUPITER 37A

JUPITER 37A

Focal length: **135mm**
Aperture: **f/3.5-22**
Aperture Blades: **12**
Minimum Focus: **1.2m**
Focus Throw: **270°**
Filter Thread: **52mm**
Weight: **410g**
Length: **90mm**
Avg. Price: **$40-90**

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The Jupiter 37A is a longer telephoto than the Jupiter 9 and the best performer when compared its 135mm sibling, the Jupiter 11. The Jupiter 37A resolves 45/30 lines per mm at center/edge, while the Jupiter 11 is only capable of 34/19. The design here is very simple and "inspired" by Carl Zeiss' Sonnar 135mm f/3.5, with key differences regarding the glass used in its making - so it is not as expensive as Zeiss' lens. A maximum aperture of f/3.5 is not impressive for a telephoto lens, but it is not bad either.

The "A" in the name stands for its ability to swap mounts for different cameras, notably Nikon and Pentax. One of the 37A's weakest aspects is its performance with backlighting, which brings intense flaring and bloom to the picture. Another common issue is having too much oil inside the barrel which, with time, spills over the aperture blades. So be sure to check for that when buying yours - or get it cleaned afterwards.

A second version, the 37AM, introduced multi-coating and is an improvement over the original design, making it more reliable and lighter all at once. The coatings make it more flare-resistant and increase contrast across the frame. The downside is that they are almost as twice as expensive as the original version.

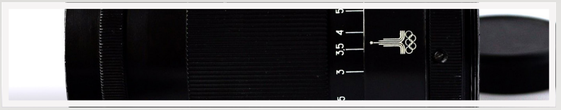
Notice the difference in the coatings between the 37AM and 37A. Photo by Radojuva.

The preset aperture works in an unusual way on this lens, as explained by reviewer Biulazz: *"turn ring at F22, pull the ring towards the body, turn to the desired preset aperture and release; use it as a normal Preset Aperture ring; then pull again and turn at F22 to cancel preset".*

USSR LENS BUYER'S GUIDE

Jupiter 37A in action. Photo by Yuri Leo.

If you are the collector type, there is a special edition of the Jupiter 37A released for the 1980 Moscow Olympic Games. It features the Olympic symbol engraving on the lens body and a super-fancy lens case with golden inscriptions. These changes are only cosmetic and there is no optical difference between this version and the others.



The 1980 Moscow Olympics special edition of the Jupiter 37.

Overall the 37A is a fairly light and compact 135mm when compared to the available alternatives, making it a useful travel lens for close ups with great, yet not mindblowing, performance.

JUPITER 6

JUPITER 6

Focal length: **180mm**
Aperture: **f/2.8-22**
Aperture Blades: **18**
Minimum Focus: **2m**
Focus Throw: **300°**
Filter Thread: **77mm**
Weight: **1470g**
Length: **139mm**
Avg. Price: **$350-450**

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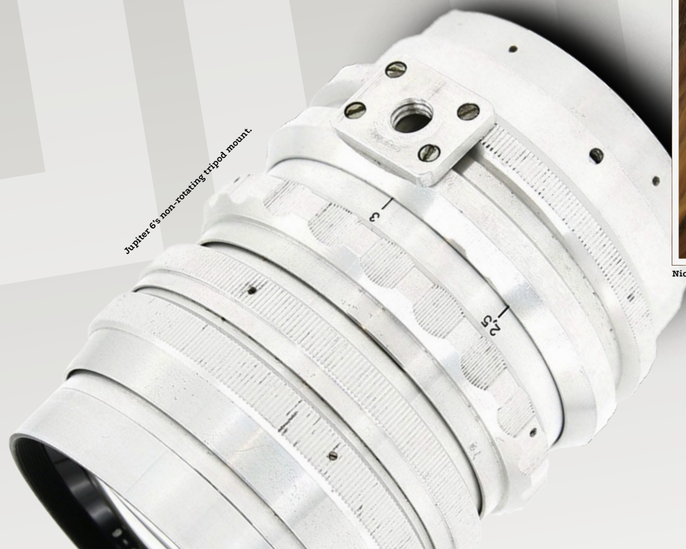
This is the lens you want to take to a alleyway brawl. With almost one and a half kilograms (3.3lbs), the Jupiter 6 is literally a tank, built entirely of glass and metal. Such construction is also the biggest downside of this fast telephoto – an uncommon treat, great for low-light scenarios.

It works for close up of tiny things too! Photo by Pierre Tizien.

Derived from Zeiss' Olympia Sonnar 180mm f/2.8, the Jupiter 6's 18 iris blades ensure round and beautiful bokeh throughout the entire aperture range. The first version of the Jupiter 6 has a silver body and features a M39 mount. The mount can be adapted to M42 through the use of a small adapter ring. The following version, Jupiter 6-2, has a native M42 mount and features an all-black body, just as heavy and sturdy as the original one.

The black version is slightly less prone to internal reflections and light bouncing inside, but both of them suffer from intense flaring when shooting towards strong lights. A properly sized hood can make a great difference in the flaring issue, but the Jupiter 6 invariably falls on the low-contrast range of the spectrum.

USSR LENS BUYER'S GUIDE



Due to its massive weight, the Jupiter 6 has an embedded tripod mount to avoid wear and tear on the camera mount. The weird part is that you cannot rotate the tripod mount to face downwards, requiring a special M42 adapter to rotate it to the right position!

Nice and smooth rolloff between in-focus and out-of-focus areas. Photo by Pierre Tizien.

JUPITER 21M

Focal length: **200mm**
Aperture: **f/4-22**
Aperture Blades: **8**
Minimum Focus: **1.8m**
Focus Throw: **300°**
Filter Thread: **58mm**
Weight: **980g**
Length: **165mm**
Avg. Price: **$70-180**

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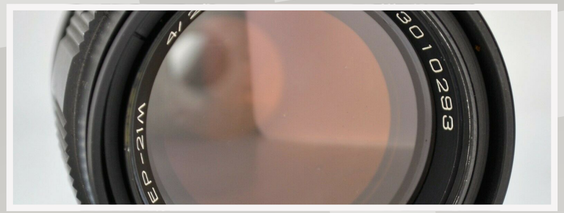
Like a punched-in version of the Jupiter 37A, the 21M delivers sharp results at 200mm and not-so-fast f/4 aperture. It is an underrated and – fortunately – undervalued lens. The main challenges of working with it are its size and weight, which can be limiting to some filming styles such as all-handheld or docu-style. The good thing about its weight is it means a strong build and endurance to the passage of time without issues.



A closer look at the Jupiter 21M.

Wider frames with a 200mm lens render subtler perspective lines. Photo by BurstMax.

Continuing to plague the Jupiter family, backlighting and flaring are two downsides of the Jupiter 21M. The lens has a built-in hood, which helps a bit. Users recommend a small modification to a reflective inner surface to improve contrast. Known to be as sharp as some Carl Zeiss and Pentacon lenses, the Jupiter 21M's sweet spot is at f/5.6, with lots of fuzziness and chromatic aberration when wide open.

The covered-up metal plate inside the rear of the Jupiter 21M for less flaring. Modification by Biulazz.

The aperture ring is not preset on this one, meaning it has clicked stops. An earlier version, the Jupiter 21A – based on a 200mm f/4 Sonnar design –, features the same replaceable mount as the 37A, but at much higher prices than the standard 21M. The 21A is more sought after because of its early production, the ability to fit different cameras and mainly because of preset aperture, which renders them click-less and ideal for video. The Jupiter 21A's aperture ring is at the front, as most preset lenses, while the 21M's is at the back.

Jupiter 21A on the left and 21M on the right. Photo by Torbjörn Boxell.

USSR LENS BUYER'S GUIDE

MIR 20M

Focal length: **20mm**
Aperture: **f/3.5-16**
Aperture Blades: **6**
Minimum Focus: **0.18m**
Focus Throw: **180°**
Filter Thread: **N/A (rear)**
Weight: **390g**
Length: **63mm**
Avg. Price: **$300-450**

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Descending from the legendary Zeiss Flektogon 20mm f/4, the Mir 20M has big shoes to fill. This lens is my choice as the widest option in this guide as it displays a rectilinear perspective without pronounced barrel distortion. There are wider Soviet lenses out there such as the Zenitar 16mm or Peleng 8mm, both of which are fisheye and deliver a completely different image from this ultra wide angle.

MC Zenitar 16mm and Peleng 8mm.

The Mir 20M is susceptible to pronounced flares and renders low contrast, vintage-looking images. This is partially to blame on its low resolving power. The multicoated versions improve contrast and color rendition but do not make the lens less vulnerable to flares. The flares are pretty neat though, a perfect rainbow circle around the light source.

Rainbow flares that match many other lenses in this guide. Photo by Felipe Saldanha.

This lens has essentially three versions*: the one shown in the profile picture; the MC Mir 20N, which goes on Nikon cameras and features multicoating; and the MC Mir 20M, which went into production in the mid 1990's and is still made today. All versions feature a large, protruding front element which is a hint for their poor field curvature.

The protruding front makes it hard to use filters on either versions of the Mir 20M, with the solution being 28mm rear-mounted filters. The Mir 20N can take 77mm filters when paired with a spacer ring so the filters do not hit the front element.

* Actually, there is a fourth version, the Mir-20 automat, for Kiev 10 and 15 SLR cameras, but they do not work for our M42 goal.

Rear mounted 28mm filters.

The Mir 20M is most appealing on a full frame camera as its wideness does not stand out when used on a S35 or APS-C body. On the downside, while the center is reasonably sharp, corner and edge performance drop fast at any f-stop and you will notice that poor field curvature, with areas at different depths coming in and out of focus.

Landscape photography: what else to expect from the Mir 20M? Photo by Felipe Saldanha.

On the subject of focus, with a slower aperture of f/3.5, focusing on the Mir 20M is a partially pointless task. It is very easy for your subject to be beyond the hyperfocal distance, as that sits at 3.75m (12ft) even when using the lens wide open. Conversely to this, the lens offers an amazing close focus range, coming down to 18cm (7in). Such close focus allows for dynamic compositions with the subject large in the frame and yet capturing much of the environment around it.

Great minimum focus distance creates compelling compositions. Photo by Felipe Saldanha.

In its latest version, the MC Mir 20M offered by Zenit, the lens features some plastic elements in its construction - unlike its predecessors - but that does not affect its performance. The biggest drawback of the modern Mir 20M is most of its inscriptions are simply painted onto the lens instead of engraved, meaning they will get wiped off with time and friction.

Zenit's MC Mir 20M.

MIR 10A

MIR 10A

Focal length: **28mm**
Aperture: **f/3.5-22**
Aperture Blades: **6**
Minimum Focus: **0.2m**
Focus Throw: **275°**
Filter Thread: **67mm**
Weight: **520g**
Length: **81mm**
Avg. Price: **$380-500**

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The Mir 10A is voted by many as the most inexpressive Soviet wide angle, which is a harsh description for it! It features an original design from the State Optical Institue, instead of being another Zeiss copy. To be honest, if you have to choose a lens to skip on this list, this is the best candidate for it. Not as wide as the Mir 20M and not as fast as the Mir 24M, the 10A is sort of "in the middle of the way".

A very Canadian scene seen through the Mir 10A.

The Mir 10A features a long focus throw, great close focus capabilities and very little distortion. These three traits combined make up the best type of composition using this lens: extremely wide close-ups.

The Mir 10A's aperture shape and its effects on flares and bokeh. Photos by Mark Holtze.

I really enjoy these wide close ups. Photo by Mark Holtze.

Although technically unremarkable, this Mir is a comfortable lens to operate and probably one of my favorite 28mm lenses ever in spite of its weight. Going against the tradition of Soviet lenses that get completely washed out when flares come up, this Mir has a fairly decent flare resistance, not being easily blasted to white.

Like the Helios 40, the Mir 10A displays a strong ninja star aperture shape when stopped anywhere between wide open or all the way closed. This gives it unique bokeh and also pretty peculiar lens flares.

USSR LENS BUYER'S GUIDE

MIR 24M

MIR 24M

Focal length: **35mm**
Aperture: **f/2-16**
Aperture Blades: **6**
Minimum Focus: **0.3m**
Focus Throw: **220°**
Filter Thread: **58mm**
Weight: **370g**
Length: **64mm**
Avg. Price: **$350-500**

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Designed at the end of the 1960's the Mir 24 is an original Soviet optical formula. Its challenging manufacture pushed actual production by more than twenty years and the the lens only went to the market in 1985! Unlike the unremarkable Mir 10A we just looked at, the Mir 24M is praised as one of the best wide angles made in the USSR.

The Mir 24M is the fastest lens in the Mir lineup, at f/2. Photo by arsenyi.shapurov.

It is important to highlight there are two versions of this lens: the Mir 24M, with our chosen M42 mount, and the Mir 24N, fit for Nikon cameras. The general consensus is the 24M has superior construction as those were still made under USSR standards, while the 24N only came to the market in 1995, after the collapse of the Soviet Union. The Mir 24N was only produced at the Arsenal plant, which featured much lower quality standards than KMZ's.

Medium close ups with a 35mm are challenging. Photo by Keita Nakayama.

The Mir 24M is a multicoated lens - there are copies without the "MC" inscription but they are still multicoated. You would be a fool to think this means the lens is particularly resistant to direct flares and backlighting. Poorly blackened pieces lead to reduced contrast and veiling glare in the presence of large light sources (such as a cloudy sky).

The frame design is pretty similar to the Helios 44M, with the focus ring at the front and a non-preset aperture ring at the back. This means the transition between different stops comes with a click as the iris moves its six blades. The small blade count means bokeh will not be perfectly round unless the lens is used wide open.

On that note, the Mir 24M has debatable performance when wide open, with good sharpness only in the center of the frame and lots of chromatic aberration as we push to the corners. You need to stop it down to f/4 or f/5.6 to completely sharpen up the image and eliminate chromatic aberration.

A peculiar aspect of the Mir 24M's design is the spacing of the focus range. As you can see in the marks in the next image, we have great control between minimum focus and 3m (10ft), but then focus quickly jumps to infinity, making it hard to get accurate focus on subjects anywhere further than close up range and closer than infinity.

The last thing to keep in mind is even though we have a full stop of additional light when compared to the next lens, the Mir 1B (37mm f/2.8), the Mir 24M displays intense vignetting towards the corners of the frame when wide open. This was one of the compromises in the design in order to achieve a maximum aperture of f/2 and cover full frame.

The sky goes pretty dark in that corner because of vignetting. Photo by Αρτεμις.

MIR 1B

MIR 1B

Focal length: **37mm**
Aperture: **f/2.8-16**
Aperture Blades: **10**
Minimum Focus: **0.7m**
Focus Throw: **270°**
Filter Thread: **49mm**
Weight: **200g**
Length: **59mm**
Avg. Price: **$80-120**

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The Mir 1B is based on Zeiss' Flektogon 2.8/35 design which was appropriated by the Soviets at the end of World War II. It is not an identical copy as the original blueprint was adjusted to comply with a different type of glass, but still retains the design of six elements in five groups.

The Mir 1B is a great match for anamorphic adapters.

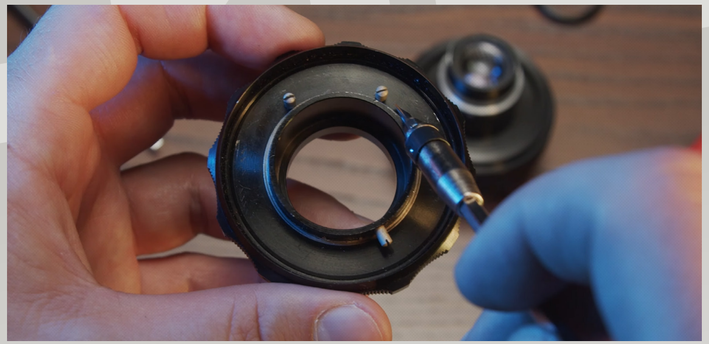
The redesign was quite successful as the Mir 1 also landed the Brussels 1958 World Fair Grand Prix along with many other lenses in this guide. Some Mir 1's display the Grand Prix inscription which makes them more expensive due to collector value. They have no optical differences though!

The Mir 1B is a great pairing to the Helios 44 as it displays slight swirls on the out-of-focus areas, yet at a wider field of view. It produces very organic and vintage looks, with reduced contrast and a great blend of softness and resolving power. The center is always sharp, but to get the edges you must stop the lens down to f/5.6 to f/8, otherwise you will notice lots of chromatic aberration and overall softness towards the corners. Bokeh is smooth and bubbly thanks to the number of aperture blades. This lens is also on the list of easy-flaring Soviet optics, with users really passionate about its flares.

A couple samples of the Mir 1B's flares. Photos by The Film People.

The main criticism to this lens is how narrow the focus ring is, and how it is positioned super close to the camera body at the back of the lens, making for cumbersome handling. The big preset aperture ring sits where you would expect the focus ring to be.

Speaking of focus, the Mir 1B has a lousy minimum focus when compared to the precedent set by the other Mir lenses. 70cm here compared to 30, 20 and 18 in the previous three lenses. Fortunately there is a simple mod you can do to half that distance down to 35cm (1ft 2in).



A little open-heart surgery to reduce the Mir 1B's minimum focus. Photo by HYPERLENS.

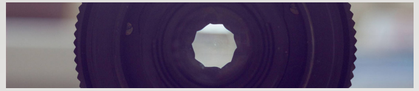
The Mir 1 has a multiplicity of names and it is useful to cover those differences. The original - Mir 1 - is almost always silver, comes in M39 mount, and is the first generation of these lenses after the war. There are multiple reports these early versions feature better mechanical construction, but single coated optics, making them less contrasty and more prone to flaring.

It is possible to find black Mir 1's, with M42 mounts as well, but these are much harder to come by.

After the Mir 1 comes the Mir 1B or 1V (B is Cyrillic for V in Latin). Seen in the profile picture, the 1B is the mass-produced version, hitting the shelves consistently for over 40 years. It is not uncommon for these to display oily aperture blades, or have loose screws inside, making a rattling sounds - I had a couple noisy ones. All models that came after have multicoated optics, which greatly improve contrast and color rendition when compared to the original Mir 1.

The uncommon Mir 1/37A.

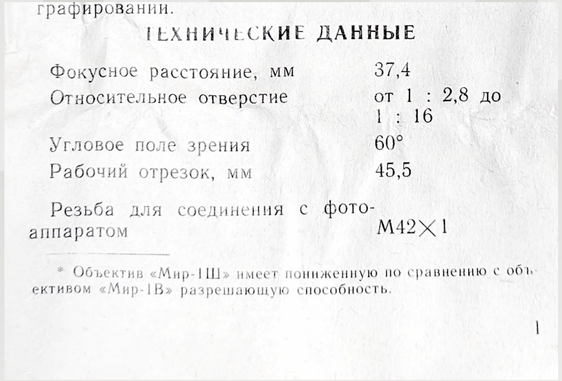
The Mir 1/37A was an attempt to modernize the design, implementing plastic pieces, a replaceable mount (that "A" code we talked about before), and a much shorter minimum focus at 24cm (10in) opposed to the original 70cm (2ft 3in).



The aperture design also changed, showcasing a ninja-star shape much like the Mir 10A and Helios 40.

Unfortunately the 37A had a very short run between 1972 and 1973, making it super hard to find and selling for ten times the price of a regular Mir 1B.

Last, there is the Mir 1Щ, or Mir 1Sh, made between 1987 and 1991, which has plenty of myths around it. I found reports of it being developed for schools and limited budget organizations, where the Щ engraving prevented it from being sold to thrift stores. It turns out the Mir 1Щ is nothing but a budget-version of the Mir 1B, as stated in the lens' manual. Soviet Lens Reviews shows the Mir 1Щ has half of the Mir 1's resolving power.



графировании.

ТЕХНИЧЕСКИЕ ДАННЫЕ

Фокусное расстояние, мм	37,4
Относительное отверстие	от 1 : 2,8 до 1 : 16
Угловое поле зрения	60°
Рабочий отрезок, мм	45,5
Резьба для соединения с фотоаппаратом	M42×1

* Объектив «Мир-1Ш» имеет пониженную по сравнению с объективом «Мир-1В» разрешающую способность.

1

First page of the Mir 1Щ's manual: explaining the difference to the Mir 1B.

Just one more sample frame with the Mir 1B. Photo by The Film People.

TAIR 11A

TAIR 11A

Focal length: **135mm**
Aperture: **f/2.8-22**
Aperture Blades: **20**
Minimum Focus: **1.2m**
Focus Throw: **330°**
Filter Thread: **55mm**
Weight: **600g**
Length: **110mm**
Avg. Price: **$280-350**

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Right next to the $350 Starter Set there is the Tair 11. I only skipped it on the set because it would mean a big price increase, but this lens is almost as popular as the Mir 1B, Helios 44-2 and Jupiter 9. The Tair 11 is the natural next step for those who feel 85mm is not long enough.

The Tair 11 has it all for a great telephoto! Featuring a preset clickless and fast aperture at f/2.8, a long focus throw, compact size, an outrageous number of aperture blades, this lens creates images unlike any other.

The most common version is the Tair 11A, which is black, has a replaceable M42 mount and reads 135mm. It is the one that has the best resolving power and most convenient design, with a built-in hood to fight off strong flares and veiling glare. There are a few other versions, though, like the uncommon and sparsely documented Tair 11-2, which is supposed to be M42 and have focal length of 133mm – not that the 2mm focal length difference would be noticeable.

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Another option for the hardcore collectors is going to the early versions, which are silver. These are also 133mm, but have minimum focus at 1.5m as opposed the 1.2m on the 11A. Some of these early versions feature the Brussels 1958 World Fair Grand Prix inscription, awarded to the Tair 11 for its outstanding technical achievement, along with other USSR lenses. Even though it is an uncommon marking, the Grand Prix units have no difference other than cosmetic from the crowd of silver Tair 11's.

If you are even more hardcore into collecting old lenses, there is the pre-series version, with #00 serials featuring the П and single coatings. These units are hard to come across and fetch much higher prices than any other versions. What makes them extra hard to find these is because many of the Brussels Grand Prix lenses have #00 serials but do not feature the blue-purple coating nor the П.

TAIR 3A

TAIR 3A

Focal length: **300mm**
Aperture: **f/4.5-22**
Aperture Blades: **16**
Minimum Focus: **3m**
Focus Throw: **360°**
Filter Thread: **72mm**
Weight: **1340g**
Length: **256mm**
Avg. Price: **$200-350**

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Awarded the 1958 Brussels World Fair Grand Prix as *"one of the most dramatic telephoto lenses"*, the Tair 3 is a lens that can do wonders when used in the right settings.

It can be soft wide open, but stopping it down to f/8 or so will not kill bokeh. Granted, 16 blades help a lot to keep the out-of-focus areas smooth and round.

From the very start of my camera life, I always wanted a 300mm. The Tair 3A was my first experience. The Tair 3A is known as the "light version" – as there are many different Tair 3's - and I am grateful I went that route. The downside is that "light version" is already 1.3kg or almost three pounds heavy and over 25cm long, making any camera look funny and small at the end of the barrel.

The most famous version of the Tair 3, however, is the 3S (or PhS), also known as the Photosniper because it came with a gun stock. To help with the *"shooter's posture"*, focus was adjusted by a knob on the underside of the lens. I strongly advise AGAINST using this ensemble out in public. The booklet and ads for the Tair 3S are truly unique.

If the whole gun stock was not enough to draw attention, you can get this lens in light grey, "Stormtrooper" style.

The light grey Tair 3s. Photo by hin_man.

The technical aspects of the Tair 3S reinforce my advice against it: the Photosniper is even heavier than the Tair 3A, at 1.6kg or 3.5lbs. Focus is cumbersome and you need creativity to rig the stock's trigger to your modern camera's shutter.

But back to the practical versions: choose between the original Tair 3 and the 3A. The advantages of the 3A over the original are a lighter build, a tripod mount that can be rotated around the lens' body and a built-in hood. The hood is a true blessing as the Tair 3 is a sucker for backlighting and glare. Later versions even added multi-coating to the mix, improving contrast and color vibrance.

300mm does a great job of compressing things together for dynamic compositions.

The original Tair 3 sometimes has a tripod mount, and sometimes also features the 1958 Grand Prix inscription (like the Tair 11, Mir 1B and MTO-3M5CA). It lacks the preset aperture though, operating through clicked stops.

Canon 5D3 + Tair 3A @ f/8.

I am always surprised by how amazing pictures with the Tair turn out. Every time I mount it on the camera I think *"I should use this lens more often!"*, but doing so requires planning and foresight. It is unwise to lug around a 300mm attached to your camera or inside your backpack...

INDUSTAR 61 L/Z

Focal length: **50mm**
Aperture: **f/2.8-16**
Aperture Blades: **6**
Minimum Focus: **0.3m**
Focus Throw: **380°**
Filter Thread: **49mm**
Weight: **225g**
Length: **59mm**
Avg. Price: **$60-90**

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The Industar 61 derives from Zeiss' Tessar design and features four elements in three groups. It was one of the few Soviet lenses to use lanthanum in its construction, a heavier glass than the traditional flint, which led to much sharper resulting images.

It is nice to show a lens that is undoubtedly sharp! Photo by Attila Terdik.

There are multiple versions of the Industar 61, including many designed for rangefinder cameras that will not work for us. When looking for yours, stick to the L/Z version, made at the LZOS optical plant . The "L" means lanthanum and "Z" means this lens is fit for SLR cameras. There are both mutlicoated (MC) and single coated (no inscription) models. The MC version delivers better contrast and colors.

The different coatings of the Industar 61 L/Z depend on when the lens was made.

USSR LENS BUYER'S GUIDE

With deeply recessed optics, the front part of the Industar's body does a fairly good job as a hood against stray light, defending the image from unwanted flares. With a slower aperture than the all-time popular Helios 44, the Industar 61 was planned as a link between the super cheap and slow Industar 50-2 (50mm f/3.5) and the aforementioned Helios 44.

Industar 50-2 (50mm f/3.5), Industar 61 L/Z (50mm f/2.8), Helios 44M (58mm f/2).

The Industar 61 L/Z has six lightly curved aperture blades which form an angular ninja star shape when stopped between f/5.6 and f/11, only becoming a hexagon again at f/16.

Industar 61 aperture shape and its effects on bokeh. Photo by Claudia G. Kukulka.

Even though its focal length is ideal for portraits - especially on crop sensors like S35 or APS-C - the unforgiving lanthanum optics do not render skin in a flattering way, making every little imperfection visible (pores, pimples, wrinkles!).

Such resolving power, combined with its super close focus at 0.3m (1ft), and an awesome focus throw of more than a full turn for refined focus control make the Industar 61 L/Z a popular recommendation as a cheap macro lens. It is commonly paired with cheap M42 macro tubes (or spacers) to achieve even closer focus at the sacrifice of infinity focus.

The Industar 61's macro potential and incredible sharpness. Photo by Vanechka Fotolub.

MTO-3M5CA

MTO-3M5CA

Focal length: **500mm**
Aperture: **f/8**
Aperture Blades: **0**
Minimum Focus: **3.8m**
Focus Throw: **310°**
Filter Thread: **77/72mm**
Weight: **1225/620g**
Length: **80mm**
Avg. Price: **$100-170**

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Moving towards the more exotic lenses in the lineup, the MTOs (acronym for "*Maksutov's Tele Ob'ektiv*") are catadioptric lenses. Designed by Dmitry Dmitrievich Maksutov, a Russian optical engineer and astronomer, this lens was awarded the *Grand Prix* in the Brussels Expo-58 world fair for technical achievement.

MTO-3M5A, the original USSR 500mm mirror lens, showcasing its awards.

Through the use of mirrors - 3M translates to Cyrilic ZM, which stands for *Zerkal'nyi* [Mirror] *Maksutov's* -, the designer was able to greatly reduce the MTO's size and weight compared to traditional long telephoto lenses. The main issue with mirror lenses is that they have no adjustable aperture, featuring a "donut" bokeh that draws plenty of attention.

Look at those donut-shaped out of focus areas. Photo by Tomas Geher.

The MTO-3M5A is the first version of this lens and even though some of these have a pretty inscription about the prize, I advise against the first version as it weighs over 1Kg. Filter threads are 77mm and it comes with an embedded lens collar for tripod mounting with two slots – one for landscape and another for portraits. A few years later the body went through redesigning and the MTO-3M5CA was released weighing half of its predecessor: 620g. The sacrifices: eliminating the lens collar and built-in hood. The 3M5CA is much smaller and easier to handle. It also takes smaller, 72mm, filters.

Both versions have issues with contrast and direct light, delivering washed out results in those conditions. The slow aperture also makes it hard to use them at any other time than when the sun is peaking in the sky. Both the 3M5A and 3M5CA can be found featuring a MC inscription on the front ring, hinting at better coatings.

In spite of low contrast contrast, both show good resolution and allow for unique shots at such great focal length. One of its main challenges for me is dealing with minimum focus at almost 4m - 12ft -, as I always tend to get closer and closer to my subjects. Depending on your camera, you might also face issues with rolling shutter, which becomes terribly noticeable at 500mm. My recommendations are for a sturdy tripod and to avoid windy locations.

At 500mm there is a great chance that the weather will influence your shots as well. If the day is too hot, for example, you will certainly be able to see heat ripples distorting the image, caused by the hot air ascending – like what you see near roads on sunny days.

In the late 80's and early 90's came the MTO-3M6A, which has the same focal length (500mm) but is 2/3 of a stop faster than the 3M5A, at f/6.3 instead of f/8. Unfortunately, these are harder to find, heavier, and even bigger than the original 3M5A, with 95mm filter threads and 4.5m as minimum focus. They are also pricier, trending just over $200.

The last offering in this range – slightly longer, at 550mm - is the MTO-500A, with f/8.5 aperture. The 500A is slower than other versions and almost as heavy as the original MTO-3M5A, weighing 1.1Kg. Produced during the 70's, the only positive aspect of this version is its reduced price compared to the others, going for around $100 – which is easily trumped by the additional shipping costs due to weight.

MTO-11CA

MTO-11CA

Focal length: **1000mm**
Aperture: **f/10**
Aperture Blades: **0**
Minimum Focus: **8m**
Focus Throw: **300°**
Filter Thread: **116mm**
Weight: **1950g**
Length: **126mm**
Avg. Price: **$280-350**

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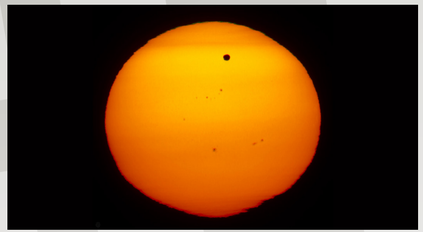
What can you expect from a lens that has a 200m focus mark before it hits infinity? The MTO-1000 is the bigger sibling of the MTO-3M5A, at 1000mm. Anything meaningful from the previous lens is doubled on this one. The first version was the MTO-1000, which weighed 3.5Kg or almost eight pounds.

Also using a mirror design, the MTO-1000 delivers donut bokeh and is much more compact than what a standard 1000mm lens would be. All members of the MTO 1000mm lens family are surprisingly sharp and allow for you to see fine detail kilometers away. There is twice as much rolling shutter as you could notice on the MTO-3M5A and filming handheld with this lens would be nothing short of irresponsible.

MTO-1000 on a Sony A7 series camera. It looks a bit silly, does it not

USSR LENS BUYER'S GUIDE



The big black dot in front of the Sun is the planet Venus. Photo by Wojciech Grzanka.

After the behemoth first version came the MTO-11 which was more than one kilogram lighter, at 2.3Kg (5lbs), and slightly slender when compared to the previous version. The MTO-11 was followed by the MTO-11CA, which is the lightest of the bunch, weighting "only" 1.9Kg (4lbs). The strenuous weight for these lenses comes from the all-metal construction plus embedded lens collar and built-in hood that are present in all available versions. Although meaning extra weight, the hood is great for controlling flares on the MTO's large front element.

The extreme focal length means this lens can actually double as a telescope – and that is quite common practice – since it can be fitted with a T-mount at the back. The extreme focal length is great for flattening perspective and making things seem much closer than they actually are.

1000mm is amazing for flattening out depth. Photo by Wojciech Grzanka.

7. RARE AND EXOTIC LENSES

On the following list are hardcore lenses that are special just for existing. I could not find much info on almost any of them besides confirming they actually exist. I thought of tracking them all while writing this guide, yet the only ones I could find during almost a year of searching were the "non-legendary" ones: the Zenitar ME1 and the Cyclop H3T-1.

Vega 13M (100mm f/2.8)

The Vega family is mostly known for lenses designed for enlargers, medium format cameras, and 16mm motion picture cameras. The ones designed for 16mm film are still a great choice for smaller sensors such as Micro Four Thirds. The Vega 13 is an exception to the family. Designed for 35mm SLRs, this is a rare, high quality 100mm lens with good low-light capabilities.

The Vega 13 comes in four different versions (M, A, H or K), but the only ones worth of note are the 13M, which features a M42 mount, and the 13A, which has a replaceable mount, but defaults to M42.

Vega 12B, 90mm f/2.8 for Kiev-6 cameras: an alternative to the Vega 13M.

Similar in construction to the Vega 12B (90mm f/2.8 for medium format), the Vega 13 had a small production run, very likely experimental, and the same design was also used on a batch of prototype cinema LOMOs. Even though I managed to find the Vega 13 on eBay once, it sold for a price I could not pay!

Cyclop H3T-1 (85mm f/1.5)

Derived from the Helios 40 design (85mm f/1.5), which draws from Zeiss' 75mm f/1.5 design, the Cyclop would be an affordable alternative to the more expensive Helios.

That was until around 2012 when someone reached the conclusion this was a *"cheap and more limited, Helios 40"*. Since then the prices have gone up and, even though still cheaper than the Helios, the Cyclop fetches about two thirds of the price of its competitor.

Used in nightvision equipment and telescopes, the Cyclop H3T-1 has no aperture blades. It also has no focus markings, coatings and hood. Looking at it, the Cyclop is an inconspicuous, black metal lens with a few inscriptions on the front ring. What makes it stand out is the same strength as the Helios 40: exceptional low-light performance and swirly bokeh.

If you want to be creative, you can cut different aperture stops using cardboard, acrylic or similar material, and place them behind the lens - very much like a Lensbaby. This will increase your depth of field and sharpness. Some versions of the Cyclop have a tripod mount, which helps balancing out the 450g of its weight on the camera.

Intense bokeh and swirls from the Cyclop H3T-1. Photo by Mathieu Stern.

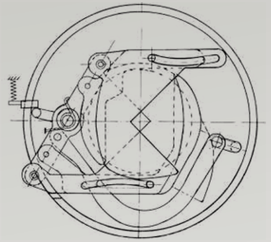
Zenitar ME1 (50mm f/1.7)

Not to be mistaken by the Zenitar M 50mm f/1.7, which has 6 aperture blades, the ME1 has two blades and a unique iris design which gives bokeh a perfect square shape. This allowed the lens to have both a lower production cost and a higher resolving power.

It used to be the kit lens for the Zenit-18 and now has a cult status because of its unusual square bokeh. Its *"optical scheme was borrowed from the Leica Leitz's Summicron 50"*, according to the review on allphotolenses. High quality glass was used in its construction, yielding powerful results on the images produced.

Square bokeh and extra sharp focus. Photo by Soviet Lens Reviews.

The unusual bokeh shape and superior materials used gave it special attention and drove the prices high. It usually goes for above $300, even though it is not a superb contender among the huge variety of 50mm lenses available nowadays (both modern and vintage). All the Zenitar ME1 has is a look that sets it apart from the rest.



Iris mechanism for the Zenitar ME1

Era 6M (50mm f/1.5)

Era lenses are naturally rare because they never lost their prototype status. According to the scarce information available, the Era lens family was designed in the early 70's by three female engineers. These lenses were mostly for reproduction purposes except the 6M, which had a native M42 mount, making it the fastest USSR 50mm available. It showcases great close focusing capabilities, with minimum focus at 0.3m. The optical design is also more complex than usual, with six elements in five groups, weighing around 430g – almost twice as a Helios 44.

I was unable to find much praise (or criticism) on its performance. Yet, I found one very technical review! I only came across one eBay listing, again, way out of my budget.

Mir 46MA (35mm f/1.4)

Do not mistake the Mir 46, which is a super fast 35mm f/1.4, with the Mir 47: an average 20mm f/2.5. The Mir 46 was part of a pre-production batch that never went into full scale production, with just a few units being made in 1980 at the KMZ plant. Besides its fast aperture, the 35mm focal length is much desired for its versatility. It can be used in multiple scenarios, from astonishing landscape images to low-distortion close ups.

The Mir 46 would come in both M42 (46MA) and Pentax K mounts (46MK and 46K), featuring multi-coating, a large number of elements (ten, in eight groups) and hefty 450 grams for a 35mm lens. I would expect unique results from it wide open. Its reported 8-blade aperture is not a bokeh monster, but could be enough for smooth out of focus areas.

The Mir 46 is one of the rarest lenses in this hard-to-find list. It was a challenge to find any links of people who actually owned it.

Based on my experience with the Mir 24M – which is amazing at fast speeds – the Mir 46 should relate to its slower sibling like the Helios 40 relates to the Helios 44: a on-steroids version, better, faster, stronger. It would also be the last piece in the USSR super speed set: Mir 36MA (35/1.4), Era 6M (50/1.5) and Helios 40 (85/1.5), in which two out of three lenses are extremely hard to find!

Zodiak 2M-2 (15mm f/3.5)

While most lenses in this list were part of experimental and small production runs, the Zodiak 2M-2 never lost the prototype status: only a handful of these were made at KMZ between 1979 and 1983. The Zodiak is a 15mm fisheye lens with 180 degrees of field of view and reasonable speed even when compared to today's fisheye lenses. We have similar field of view and speed in considerably smaller designs such as the Canon EF 8-15mm f/4L, Rokinon Cine 8mm T/3.2 and Sigma 8mm f/3.5, which it makes me wonder what kind of results the 103mm front diameter delivers for the Zodiak!

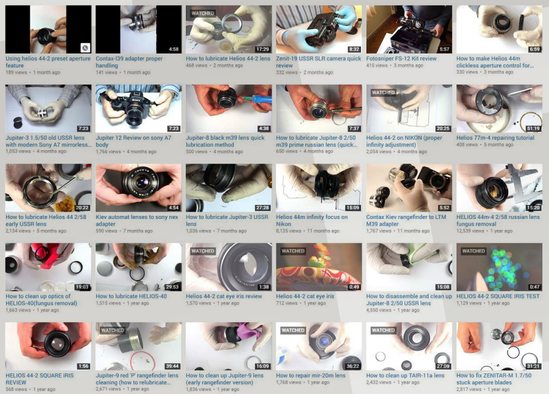
Zodiak 2M-2 and MC Arsat 30/3.5 side by side comparison.

Some people claim the Zodiak 2M-2 is the same as the Arsat 30/3.5 which features a Pentacon Six mount for medium format. To strengthen this theory, the Arsat originally went as a Zodiak lens, and both Zodiak 2M-2 and MC Arsat 30mm f/3.5 share the same speed, field of view, and rear 28mm filters. The Arsat is much easier to find, but still reaches high prices, between $250 and $450.

8. RECOMMENDED SELLER: RETRO FOTO HOUSE

RETRO FOTO HOUSE™

One of the more reliable sources I found for all kinds of information related to Soviet optics is the YouTube channel Retro Foto House kept by Roman Bibik. His channel has plenty of videos on how to fully disassemble and put together many of the most common lenses like Helios 44, Helios 40, Tair 11A, Mir 20M, among others.



This is the kind of content you find at Retro Foto House's YouTube channel.

Roman also keeps an eBay store for selling good quality USSR lenses, from which I bought several lenses in the past and highly recommend his items and service! I have exchanged several emails with Bibik to verify data for this guide and he has proven to be a very knowledgeable person in USSR photographic equipment.

9. GUIDES, TUTORIALS AND MODIFICATIONS

One of the greatest advantages of USSR lenses is how simple their construction is, to the point that pretty much anyone with average tools is able to open a lens up and either try to fix it or tweak it to their own will.

The same way it is easy to get a bad copy of these mass produced lenses, it is usually just as easy to open the defective lens up and try to get it to work properly. Sometimes one of the elements is flipped the wrong way, or the focus grease has hardened with the passage of time, or oil has dripped onto the aperture blades, or some part of it feels wonky and loose. All of those fixes are pretty simple to execute!

Finding information on disassembly is usually easy (look up the name of the lens + disassembly), but the guides are scattered all over the internet and each of them was written or filmed in a different way - some with more depth, others just a quick overview. In the following page is a good disassembly collection for many lenses in this guide

This could be your workbench any day moving forward from now.

MIR 20M

MIR 1

INDUSTAR 61

HELIOS 44

HELIOS 40

JUPITER 9

JUPITER 37

JUPITER 21

TAIR 11

TAIR 3

MTO-500

MTO-1000

Now that you are well versed in disassembling and reassembling lenses, how about some modification ideas to customize their look even further? Coming up are some of my favorite techniques, most of which are fairly simple to execute.

HELIOS 44-2 OVAL APERTURE

I am a big enthusiast of anamorphic lenses and adapters. One of the main differences between a regular lens versus an anamorphic lens is the shape of bokeh. While a normal (spherical) lens has a circular shape for its out of focus highlights, anamorphics make oval-shaped bokeh.

Thanks to a lot of advice and pioneer experimenting from Amir Safari, I learned how to make my own aperture discs and how to take the Helios 44 apart to fit the disc in there. These alternative apertures shape the way the out of focus areas look. By letting in light through an oval shape, bokeh will be equally oval. Not real anamorphic, but anamorfake enough to convince untrained eyes. The first test subject was the Helios 44-2, and I already repeated the modifications for many other Soviet lenses.

You can watch the full video tutorial for the Helios 44-2 here, and order oval apertures here. If you want to try the process for other USSR lenses, you can find several tutorials in this playlist on my channel. You can just as easily order discs from my store linked above.

Notice the oval bokeh on the background!

HELIOS 44-2 EXTREME MOD

After the oval bokeh mod, I came across a technique to drastically change the way the Helios 44's images look by polishing the glass elements and painting some of the lens' inner surfaces. This led me to develop a tutorial combining the polishing with the anamorphic look for intense results. You can check out the step-by-step here, as well as the resulting footage.

Fully modified Helios 44-2 with amber tint, polished glass and focus gear for video.

A sample frame of how moody this mod can make your footage. Yes, this has extra flares.

I was also able to expand these mods onto the $350 Starter Set, modifying both the Mir 1B and Jupiter 9 for a set of unique looking lenses through creative painting and polishing.

Fully modified set of primes with polished glass for decreased contrast and glow, oval semi-transparent apertures and anamorphic flare lines.

HELIOS 44-2 SQUARE APERTURE

You already learned about the exotic and expensive Zenitar ME1, the 50mm f/1.7 lens with a square aperture and unique bokeh. With the Helios 44 being so inexpensive, it is no surprise someone decided to modify its aperture blades to make it square-ish.

If you are a hardcore tinkerer, here is the tutorial on how to do it yourself. It is not an easy nor fast process, but can still be done by determined and skilled individuals. This is one of the most detailed tutorials I have come across, explaining every little step of the process with photos and text. At the end of it you will have a Helios 44 lens that turns bokeh into little squares, and, unlike the previous oval aperture modifications, it can be stopped down to match available light.

HELIOS 44-2 CAT EYE APERTURE

Are all of these modifications for about the Helios 44-2? Almost all, I think. That is what you get by having a lens that is so cheap, yet reliable. This modification was brought to light by Roman Bibik, mentioned earlier in this section. I also did a video review as it is similar to the anamorphic oval aperture, with the advantage of being able to stop it down without taking the lens apart every time.

I do not have a tutorial of how to do it, though. Roman Bibik from RetroFotoHouse was the only one I have seen to offer this modification.

HELIOS 44-2 EXTREME SWIRLY BOKEH

Much simpler than the previous modifications, this tutorial by Viktor Pavlovic aims at enhancing the most sought-after trait of the Helios 44: the swirly bokeh. It is said that this procedure makes the Helios' images look like a Petzval lens thanks to the emphasis to the center of the frame, with swirls all around it and a soft focus texture.

Helios 44-2 with flipped front element. Hard to tell the difference just looking at it.

I feel the effect is too intense and I am not a fan of the soft focusing, but I can see the value of its uniqueness – especially when you factor that a New Petzval 58 is almost $800 and a Helios 44 can be found for $50.

A sample of the reverse Helios 44. Photo by Noe Alonzo.

MIR 1B REVERSE GLOBULAR

To break the Helios' streak of modifications, this one is by vintage lens expert Mathieu Stern and it shows how to turn your Mir 1B into a "reverse globular lens". It is very easy to see how the modification affects the resulting images by keeping the center relatively sharp and powerfully stretching out the edges of the picture.

Mir 1B "reverse globular". Photo by Mathieu Stern.

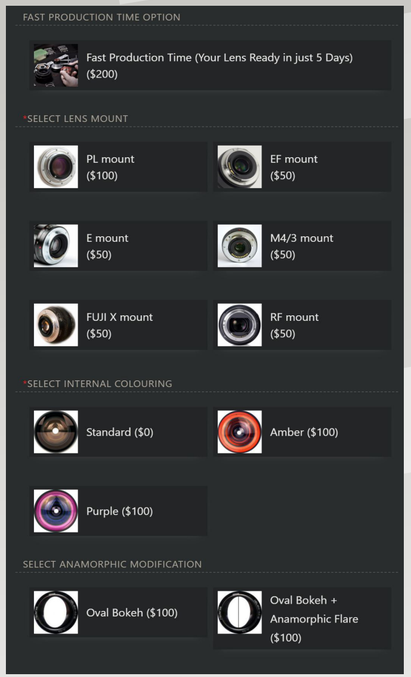
The process is basically the same as the Helios 44-2 Extreme Swirly Bokeh we just looked at. You remove the front element, flip it inside out and lock it back into place.

10. IRONGLASS LENSES

If you are living under a rock and never heard the name before, IronGlass is a company in Ukraine that sells modified Soviet lenses. They started making their anamorfakes after I posted the first few tutorials for modding USSR glass and they have taken it much further than simple modifications, developing full rehousings for many lenses mentioned in this guide. You can orders either on their eBay store or on their website.

A cine modded silver set from IronGlass.

Here is a peek under the lens caps. You can choose different tints and styles.



IronGlass offers several focal lengths: 20mm, 28mm, 37mm, 58mm, 85mm, 100mm, 135mm and 180mm. A range which can capture any scene. You can get them in sets or as individual lenses, choosing awesome customizations for lens body color, flares, oval bokeh and mount.

Advertised as cine modded lenses, they come with focus and iris gears as well as declicked apertures. The lenses also have custom backs and cine-style front rings, with 77mm threads and 80mm outer diameter.

Why is IronGlass so special? Because you can just throw them on your camera and start filming anamorphic-like footage (or anamorfake, as I call it) that has huge production value for a fraction of the cost of an actual anamorphic setup!

I have been experimenting with USSR lenses for almost ten years now and the one constant through my experience is you can never know what to expect in terms of image quality and performance. The lenses I got from IronGlass were in tip-top shape and sharper than the vast majority of the other USSR glass that passed through my hands.

Samples from Ironglass Cine Modded set with amber tint and blue flares.

USSR LENS BUYER'S GUIDE

LENS REHOUSING

Soviet still lenses always come under fire for being unreliable and lacking in terms of build quality. They are not the type of lens a cinematographer would be comfortable taking to a client shoot. They are small and look fragile, plus a lot of copies have bad performance. At the same time, they have amazing personality and have been mesmerizing the internet since someone mounted them on a digital camera.

Rehoused Helios 44-2 on a RED for high end productions. Photo by Evan Falbaum.

In partnership with Alan Besedin and Vintage Lenses For Video, IronGlass started developing full rehousings of these Soviet lenses. The Cine versions of the lenses come in EF or PL mount, feature geared focus and iris rings, no more preset aperture, and a standard 95mm outer diameter for the front, making it compatible with several clip-on matteboxes while also giving you 86mm filter threads. The construction feels really solid at 1.5kg and the rings offer the perfect amount of resistance.

During the rehousing process IronGlass also tunes up the optical performance of each lens, bringing them to top quality and delivering the best possible images for Soviet glass.

The original rehousing, VLFV Cine Edition.

The rehousings are pricy when compared to the original cost of each donor lens, but just not that pricy compared to the cine-lens league. Most other rehousing companies will say IronGlass' prices are mad cheap even though between $1000-2500 per lens.

The next thing that helps justifying their price is you can order matching lenses to complete the set such as the Mir 10A, Jupiter 9, and others, giving you identical focus and iris placements across the set - a very valuable trait of a well planned mechanical design.

A moody frame taken with the rehoused VLFV Helios 44-2.

The first version of the rehousings uses some of the original mechanics from the donor lenses. To improve on that, IronGlass developed a MkII set which built the mechanics from the ground up, relying solely on the glass of the original lenses. These look even more like standard cine lenses, yet cost more than twice than the original rehousings.

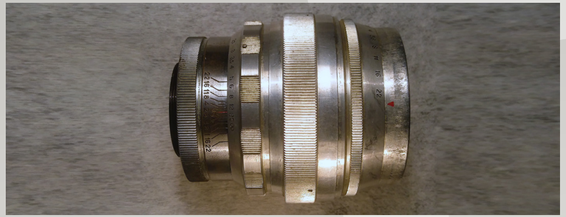
Mir 20M rehousing MkII.

11. SERVICING USSR OPTICS

As I mentioned before, over the course of a year I tracked down and overpaid for a silver, red П set of Mir 1, Helios 44, Jupiter 9, Helios 40 and Tair 11. I really liked the look they delivered, but the lenses were not at the top of their game, both optically and mechanically.

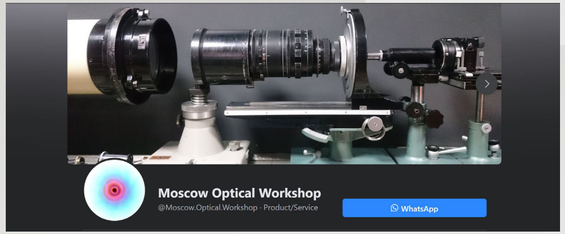
My top USSR set before servicing.

They beat my previous Soviet glass in performance yet their focus marks were off, the adapters were unreliable, the bodies looked old and yellowed, I could see dust inside the lenses, focus was not smooth and even across the range. Plenty of problems.



Notice how yellowed and grungy this Helios 40 looks.

So I went looking and found someone with a fitting background to work on them: Ilya Volkov from Moscow Optical Workshop. Actually, Ilya found me because of my videos and articles involving Soviet lenses. It is no coincidence Ilya is also the writer compiling the Russian | Soviet | USSR Lens Survival Guide.

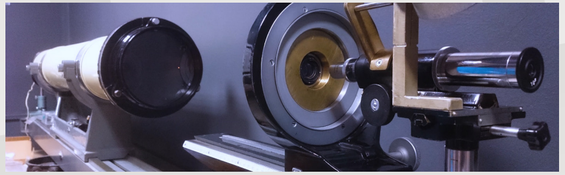


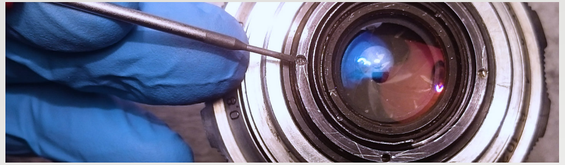
You can reach Ilya through Facebook, Instagram or WhatsApp.

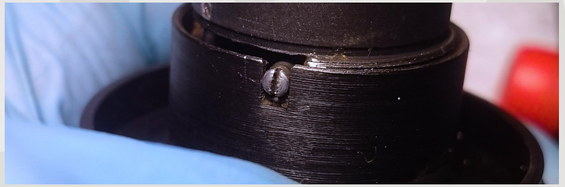
I sent him my lenses and over several months I watched him work. During that time I received constant updates, queries and comments on the process. It turned out that earlier Soviet lenses really went through a higher quality control. Mine had several bits and pieces intended to better them withstand the passing of time, bits which are not present in later models.

I also learned a handful of elements had bigger issues like haze, fungus and scratches, but Ilya was brilliant enough to source new ones that matched coatings and optics. I got constant updates through WhatsApp messages and emails that always gave me the impression that Ilya was having a blast doing this work.

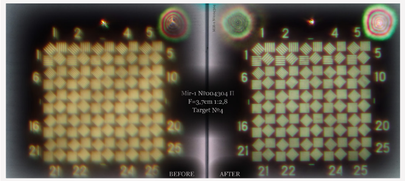
At the end of the process Volkov shipped back the lenses and sent me a long and detailed report of all the work done in each lens, as well as collimation comparisons of before/after servicing and so many photos of the process. These are just a few samples of what I got.







The lenses are now collimated to EF mount. They received custom-made aluminum focus and iris gears and polished silver shine, restoring their original color. All the engraving has been repainted and Ilya managed to modify the preset aperture to work like a cinema lens.

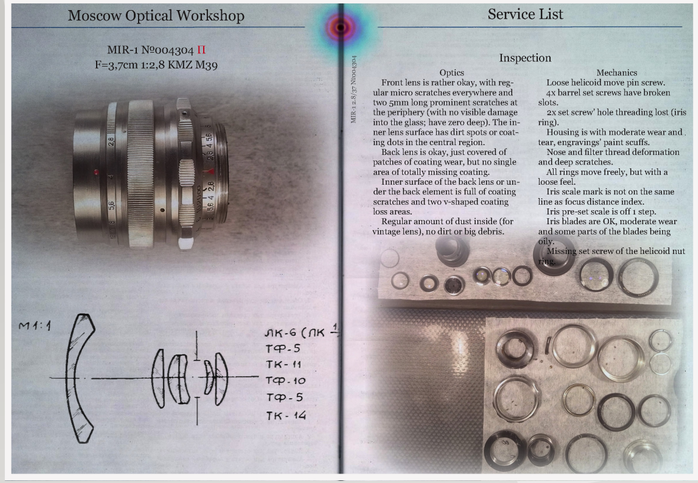


Optical bench comparison of before and after servicing, Mir 1.

These lenses went to Russia looking like ugly ducklings and came back as beautiful swans. I have had lenses serviced by a handful of folks so far and no other experiece made me learn as much about my lenses as working with Ilya.

The detailed report of the work done is the cherry on top. When I asked Ilya about why he would write such extensive documents, he said that vintage lenses should be treated like vintage cars: with detailed sheets of all the work done to them, keeping track of the historical value of those items and the upgrades they went through.

The finished project. That same Helios 40 at the front.



Mir 1 partial work report. There is even more info beyond this.

If you have Soviet glass you love and think could be better than it is, Ilya at Moscow Optical Workshop is the person for the job. I can guarantee you will learn a thing or two about your lenses and they will get way more use than before once they return.

12. THE NEW ZENITAR LENSES

Back in 2016, Russia's organization for development, production and export of hi-tech products, Rostec, surprised the camera world by announcing they would start making Zenit cameras along the same lines as Leica's. That meant high end, luxury items. Such statement makes sense since the KMZ production plant is part of Rostec's assets, and that is where the original Zenit cameras were made up until 2004, when they stopped production of all SLR bodies.

The new Zenitar lenses aimed to compete with the current batch of cheaper, yet fast and high quality primes pouring out of SLR Magic, TT Artisan, 7artisans or ZY Optics, with their f/1.2 and f/0.95 lenses for mirrorless systems.

50mm f/1.2 Zenitar for Canon EF. Photo by Denis Gavrilov.

The biggest fuss was about the super fast 50m and 85mm f/1.2 primes, but Zenit announced a big round-up of lenses, covering many focal lengths from 8mm to 135mm. According to Denis Gavrilov, author of the article linked above and someone who got to play with some of the lenses in person, most of the designs and some prototypes date decades back and it would be pointless to not use them when KMZ is perfectly capable of producing and distributing the lenses.

85mm f/1.2 Zenitar for Canon EF. Photo by Sergei Samsonov.

Several years have passed and the best we got for those promised luxury cameras was the limited-run Zenit M. Some lenses did come out though, a few of them worth of note here, some even as a warning! The 85mm f/1.2 Zenitar-N never showed up, as well as the unnamed 50mm f/1.2 shown above. There are also rumors about a 35mm f/1.0 for Leica M, which would go with the Zenit M camera.

The Zenit M and Zenitar 35mm f/1 at Photokina, by CineD.

What we can actually buy out of all the promises are: the Zenitar 50mm f/0.95 for Sony E-mount; the MC-Zenitar 50mm f/1.2 S for Canon EF and Nikon F, although only covering APS-C or S35 sensors; and the Zenitar 85mm f/1.4 which lacks the personality of the Helios 40-2. Zenit took their time selling these only in the Russian market, which made information about the lenses very hard to get, but now you can order them at B&H Photo, if you really want to do so.

I have not played with these lenses in person, but based on what I learned researching for this chapter, I think they are all a pass for me. The old stuff seems better and cheaper.

12a - Zenitar 50mm f/0.95 for Sony E

Zenitar 50mm f/0.95 on Sony A7rII. Photo by Bastian K.

The fastest one in the bunch, this 50mm f/0.95 hit the market with great expectations. Its competitors, the Mitakon Speedmaster and TT Artisans, also at 50mm f/0.95 were already established in the E-mount ecosystem as the Zenitar came in with a much larger and heavier build, along a heftier price.

This arc flare is a real eye catcher. The issue starts when you can't get a picture without them showing up! Photo by Bastian K.

In a review of this Zenitar, Bastian K. expresses severe disappointment in the lens' performance and design in almost every aspect. Conversely, in the Phoblographer's review of the same lens, there is a lot of praise for the same things Bastian criticized. Reading both of them, I am more inclined to side with Bastian, as the praise given is not uncommon among all other Soviet glass we already looked at in this guide, at a much lower price and lighter construction.

The 14 aperture blades have an unusual L shape, making it seem that there are actually two concentric aperture mechanisms. The lens also displays pronounced pincushion distortion aross the frame which needs strong adjustment in post.

50mm f/0.95 Zenitar for Sony E mount.
Photo by Sergei Samsonov.

Expect to see here what we already saw everywhere else. Weak resistance to flares, especially from direct light hits, soft performance wide open, and a lot of field curvature, making your scene seem to wave in and out of focus across the frame.

You can still make good images with a poorly performing lens. Photo by Bastian K.

From the scarce material online, we can see tons of chromatic aberration, blooming highlights and astigmatism when using the lens wide open. Reviews mention performance improves considerably when stopped by f/2, but then why carry around a 1.1kg, $1250 lens around, when you can carry a 200g, $50 Helios 44-2?

12b - MC-Zenitar 50mm f/1.2 S

Designed to cover APS-C or S35 sensors, this super fast 50mm also struggles to find a place in the market. At the price of $450, it competes with Canon's 50mm f/1.4 which covers full frame and has autofocus (not that I care for it). Zenit's only advantages in this fight would be the manual aperture and the lens' personality.

With nine aperture blades, bokeh is smooth through the iris range, and a fair word to define this lens would be "smooth", or more like "soft". As we can see in Ryan Cooper's review of the Zenitar, the center lacks sharpness when the aperture is wide open, and even stopped down the corners are still blurry.

With a full metal and glass body plus fancy packaging, this Zenitar conveys confidence through established values since too-light lenses are perceived as of lower quality. The short focus throw (180°) makes it challenging to nail focus, especially when combined with the soft touch of the images and the razor thin depth of field.

Cashing in on USSR optics' most established visual trait, the lens displays light swirls on bokeh. This appeals to modern-day photographers who are not used to the gems from the past that we have been contemplating in this in this guide.

12c - MC-Zenitar 85mm f/1.4

While the Zenitar 50/1.2 S took advantage of what we would expect from Soviet glass, the 85mm goes the other way, treading the line between vintage and modern.

Covering full frame, the Zenitar 85mm f/1.4 actually comes marginally cheaper ($460) than Canon's EF 85mm f/1.8 ($500). Unlike the 50mm which can be defined as "soft", this one sharpens up alright from f/2 and slower, with great center to edge resolution, vignette control and management of chromatic aberration.

There's a subtle hint of swirly bokeh, but this lens performs decidedly different from the Helios 40-2 (85mm f/1.5) also offered by Zenit. Do not be fooled by the claims of great sharpness though. This is still a super fast 85mm for full frame. Your depth of field will be nothing more than milimeters when in portrait range if using the lens wide open!

The Zenitar 85mm f/1.4 renders out of focus areas somewhat similarly to the Helios 40-2. Photo by Ryan Cooper.

ABOUT THE AUTHOR

I am Tito Ferradans and I am a Brazilian living in Vancouver, Canada.

For the past several of years, I have focused on sharing knowledge about anamorphic filmmaking.

My goal is to help people using anamorphics achieve a more cinematic result in their projects without having to break the bank.

I have been fascinated by USSR lenses for almost as long as I have been into anamorphic lenses, having tried hundreds of different copies and pushing this look onto many of my projects.

Contact me

If you want to get in touch, contact me at ferradans@gmail.com. And remember to subscribe to my YouTube channel for more Soviet lenses and anamorphics.

Feel free to drop me a line if you have questions or feedback about this guide or my work with anamorphics. It might take me a bit to reply, so be patient! :)

Good luck on your creative endeavors, and please send me links of projects that came to be from the tips you found in this guide.

Tito.